

COMPLETE BEST SIMPLIFIED

HERITAGE BASED SYLLABUS HANDOUT BIOLOGY

FOR FORM 3

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"SIMPLIFIED FOR
BETTER UNDERSTANDING,
EXCELLENCE IN RESULTS."



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BUILDING STRONG FOUNDATIONS FOR A BRIGHTER FUTURE



Topic: Safety in the Laboratory

Learning objectives

By the end of the lesson, learners should be able to:

- identify causes of accidents in the laboratory
- outline laboratory safety rules
- perform fire drills correctly
- memorise the local emergency numbers
- make a fire guard around the laboratory
- demonstrate the use of the First Aid Kit

1. Introduction

A biology laboratory is a place where practical work is done using specimens, chemicals, glassware, heat, electricity, and living organisms. Because of these materials, accidents can happen if learners do not follow safety rules.

Laboratory safety means all the precautions taken to prevent accidents and to protect people, equipment, and the environment.

2. Causes of accidents in the laboratory

a) Fire

Fire may start because of:

- careless handling of flames
- leaving burners on
- placing flammable materials near heat sources
- gas leaks
- electrical faults

Effects of fire:

- burns
- smoke inhalation
- destruction of equipment and buildings

b) Fumes

Fumes are harmful gases or vapours produced by chemicals or burning substances.

Causes:

- heating chemicals
- mixing substances incorrectly
- working in poorly ventilated rooms

Effects:

- coughing
- dizziness
- eye irritation
- breathing problems

c) Acids and strong bases

Acids and bases are corrosive. They can burn the skin, damage eyes, and destroy clothing.

Accidents happen when:

- chemicals are spilled
- containers are not labelled
- learners smell or taste chemicals
- chemicals are poured carelessly

d) Handling of micro-organisms

Micro-organisms may cause infections or diseases if handled wrongly.

Accidents happen when:

- cultures are opened carelessly
- hands are not washed after practical work
- contaminated materials are not disposed of properly
- instruments are not sterilized

e) Improper handling of apparatus

Glassware and instruments can cause injury when used wrongly.

Examples:

- broken test tubes cutting the skin
- microscopes falling and breaking
- sharp tools like scalpels causing cuts
- wrong use of measuring equipment leading to spills

f) Electricity

Electricity can be dangerous in the laboratory.

Accidents happen when:

- wires are exposed
- equipment is handled with wet hands
- sockets are overloaded
- switches are used carelessly

Effects:

- electric shock
- burns
- fire
-

3. Laboratory safety rules

These rules help to prevent accidents and keep everyone safe:

1. Do not run or play in the laboratory.
2. Always listen to the teacher's instructions.
3. Do not eat, drink, or chew in the laboratory.
4. Wear a lab coat and other protective clothing when necessary.
5. Tie long hair back and avoid loose clothing near flames.
6. Do not taste or directly smell chemicals.
7. Never point test tubes at yourself or others when heating.
8. Handle glassware carefully and report broken glass immediately.

9. Wash hands after practical work, especially after handling specimens.
10. Use forceps, tongs, or gloves where necessary.
11. Keep benches clean and dry.
12. Label all specimens and chemicals clearly.
13. Do not touch electrical appliances with wet hands.
14. Report all accidents, spills, and broken equipment at once.
15. Return apparatus to the correct place after use.
16. Dispose of waste materials properly.
17. Do not work alone in the laboratory.
18. Know the location of fire extinguishers, first aid kit, exits, and emergency numbers.

4. Fire drills

A fire drill is a practice exercise that teaches learners and staff what to do during a fire emergency.

Importance of fire drills

- helps people leave the building quickly and safely
- reduces panic during real fire emergencies
- teaches the correct escape route
- helps identify weaknesses in the emergency plan

Fire drill procedure

1. When the fire alarm sounds, stop all work immediately.
2. Switch off gas and electrical equipment if it is safe to do so.
3. Leave belongings behind.
4. Walk quickly, do not run.
5. Use the nearest safe exit.
6. Stay low if there is smoke.
7. Move to the assembly point outside the laboratory.
8. The teacher takes the register and checks that everyone is safe.
9. Do not return to the building until instructed to do so.

Teacher's note

Fire drills should be practiced regularly and under supervision.

5. Local emergency numbers

Learners should memorise the emergency numbers used in their area for:

- fire brigade
- police
- ambulance
- poison centre, if available
- school emergency contact

How to teach this

- write the numbers on a classroom chart
- let learners copy them into notebooks
- use songs, repetition, or flashcards
- practise emergency calling as a role play

Important point

Learners should only make emergency calls in a real emergency or when instructed by the teacher.

6. Making a fire guard around the laboratory

A fire guard is a cleared protective space around a building that helps prevent fire from spreading to the laboratory.

Purpose of a fire guard

- stops grass or weeds from carrying fire to the building
- reduces the spread of bush fires
- protects the laboratory and nearby materials

How to make a fire guard

1. Clear dry grass, leaves, and rubbish around the laboratory.
2. Create a bare strip of land around the building.
3. Keep the area regularly clean and free from fuel for fire.
4. Do not store flammable materials near the outside walls.
5. Report any fire hazard immediately.

Safety note

This activity must be done under teacher supervision and according to school rules.

7. Demonstrating the use of the First Aid Kit

A First Aid Kit contains materials used to give immediate help to an injured person before medical treatment is available.

Common contents of a First Aid Kit

- sterile gauze
- cotton wool
- bandages
- adhesive plaster
- antiseptic solution
- scissors
- gloves
- burn dressing
- safety pins
- first aid guide
- triangular bandage
- soap or hand sanitizer, where appropriate

Uses of First Aid Kit items

- **Bandages:** cover wounds and hold dressings in place
- **Gauze:** protect cuts from dirt
- **Antiseptic:** clean wounds and reduce infection
- **Plasters:** cover small cuts and grazes
- **Gloves:** protect the helper from infection
- **Scissors:** cut bandages or tape
- **Burn dressing:** cover burns

Basic first aid actions

- remain calm
- inform the teacher immediately
- assess the injury
- use the appropriate First Aid Kit item
- do not move seriously injured persons unless necessary

- send for medical help where needed

Examples of first aid in the laboratory

- **Cut from broken glass:** clean the wound, apply antiseptic, cover with sterile gauze and bandage
- **Small burn:** cool under running water, cover with clean burn dressing, seek help
- **Chemical splash in eyes:** flush with plenty of clean water immediately and call for help
- **Minor nosebleed:** sit upright and apply pressure to the nose

Important safety rule

Only trained persons should give first aid. Serious injuries must be referred to a health professional.

8. Summary

Laboratory accidents can be caused by fire, fumes, acids and bases, micro-organisms, improper handling of apparatus, and electricity. Safety can be improved by following laboratory rules, practising fire drills, knowing emergency numbers, making a fire guard, and using the First Aid Kit correctly

9. Evaluation questions

Short questions

1. Define laboratory safety.
2. State four causes of accidents in the laboratory.
3. Give five laboratory safety rules.
4. What is a fire drill?
5. Why is a fire guard important?
6. List four items found in a First Aid Kit.

Structured question

Explain how a learner should react if a fire breaks out in the laboratory.

Possible answer points

- stop work immediately
- switch off gas/electricity if safe
- alert teacher
- leave belongings
- exit calmly through nearest door
- go to assembly point
- wait for instructions

10. Practical classroom activities

Activity 1: Safety rule poster

Learners work in groups to make posters showing laboratory safety rules.

Activity 2: Emergency drill

Practise a fire drill and ask learners to demonstrate correct movement to the assembly point.

Activity 3: First Aid Kit identification

Show learners the contents of a First Aid Kit and let them state the use of each item.

Activity 4: Hazard spotting

Learners observe a picture of a laboratory and identify unsafe practices.

11. Key vocabulary

- laboratory
- accident
- fumes
- corrosive
- micro-organisms
- fire drill
- emergency
- fire guard
- first aid
- antiseptic
- apparatus
- safety rules

Branches of Biology

Learning objective

By the end of the lesson, learners should be able to identify and explain the main branches of biology.

1. Meaning of Biology

Biology is the branch of science that deals with the study of living things. It helps us understand how plants, animals, and microorganisms live, grow, reproduce, and interact with their environment.

Because living things are many and complex, biology is divided into different branches.

2. Branches of Biology

1. Cytology

Cytology is the study of cells.

A cell is the basic unit of life. Cytology helps us understand:

- the structure of cells
- the functions of cell parts
- how cells divide and grow

Importance:

It helps in understanding body growth, diseases, and heredity.

2. Anatomy

Anatomy is the study of the structure of living organisms and their body parts.

It looks at:

- organs
- tissues
- body systems
- the arrangement of body parts

Example:

Studying the bones, muscles, heart, lungs, and other organs in the human body.

Importance:

It helps learners understand how the body is built and how organs are arranged.

3. Physiology

Physiology is the study of how living things and their body parts function.

It explains:

- how the heart pumps blood
- how leaves make food
- how the lungs help in breathing
- how the kidneys remove waste

Importance:

It helps us understand how living organisms work.

4. Ecology

Ecology is the study of the relationship between living organisms and their environment.

It focuses on:

- plants and animals
- weather and climate
- soil, water, and air
- how organisms depend on one another

Example:

How frogs live in wet areas, or how plants depend on sunlight and water.

Importance:

It helps us protect the environment and understand ecosystems.

5. Genetics

Genetics is the study of heredity and variation.

It explains:

- how traits are passed from parents to offspring
- why children resemble their parents
- why individuals are different from one another

Example:

Eye colour, hair type, and blood group can be inherited.

Importance:

It helps in understanding inheritance, breeding, and genetic diseases.

6. Biotechnology

Biotechnology is the use of living organisms or their products to make useful things for humans.

Examples:

- making yoghurt using bacteria
- making bread using yeast
- producing medicines such as insulin
- improving crops through genetic methods

Importance:

It helps in medicine, agriculture, food production, and industry.

7. Microbiology

Microbiology is the study of very small living organisms called micro-organisms.

These include:

- bacteria
- fungi
- protozoa
- some algae and viruses

Importance:

It helps us understand:

- diseases caused by microbes
- food spoilage
- decomposition
- useful microbes in food and medicine

8. Zoology

Zoology is the study of animals.

It deals with:

- animal structure
- animal behaviour
- animal reproduction
- animal classification

Importance:

It helps in understanding domestic animals, wild animals, and animal conservation.

9. Botany

Botany is the study of plants.

It includes:

- plant structure
- plant growth
- plant reproduction
- plant nutrition
- plant classification

Importance:

It helps in agriculture, food production, and environmental conservation.

3. Summary table

Branch	What it studies
Cytology	Cells
Anatomy	Structure of living things
Physiology	Functions of body parts
Ecology	Relationship between organisms and environment
Genetics	Heredity and variation
Biotechnology	Use of living things to make useful products
Microbiology	Micro-organisms
Zoology	Animals
Botany	Plants

4. Why the branches of biology are important

The branches of biology help us to:

- understand living organisms better
- improve health and medicine
- increase food production
- protect the environment
- study diseases and their prevention
- improve farming and livestock production

5. Simple classroom examples

- **Cytology:** looking at onion cells under a microscope
- **Anatomy:** studying the parts of a flower or human body organs
- **Physiology:** learning how breathing occurs
- **Ecology:** observing organisms in a pond or garden
- **Genetics:** studying inherited traits in families
- **Biotechnology:** making yoghurt or bread
- **Microbiology:** observing mould or bacteria
- **Zoology:** studying insects, fish, birds, or mammals
- **Botany:** studying roots, stems, and leaves
-

6. Revision questions

1. Define biology.
2. What is cytology?
3. State the branch of biology that studies animals.
4. Which branch deals with heredity?
5. Name two uses of biotechnology.
6. What is ecology?
7. Give two examples of micro-organisms.
8. Differentiate between anatomy and physiology.
9. State two importance of botany.
10. Explain why biology is divided into branches.

7. Key vocabulary

- cell
- organism
- heredity
- environment
- micro-organism
- structure
- function
- biotechnology
- anatomy
- physiology

8. Memory aid

To remember the branches:

Cytology

Anatomy

Physiology

Ecology

Genetics

Biotechnology

Microbiology

Zoology

Botany

Careers in Biology

Learning Objective

By the end of the lesson, learners should be able to:

- List biology-related careers.
- Describe the work done by professionals in biology-related careers.
- Appreciate the importance of biology in career development.

Introduction

Biology is not only the study of living organisms but also a subject that prepares learners for many careers. Knowledge of biology is important in health, agriculture, environmental management, research, and food production.

A career is a profession or occupation that a person chooses and trains for in order to earn a living and serve society.

Biology-Related Careers

1. Medicine

Medicine is the science and practice of diagnosing, treating, and preventing diseases.

Professionals in Medicine

- Doctors
- Nurses
- Pharmacists
- Dentists
- Physiotherapists
- Medical Laboratory Scientists

Duties

- Diagnose diseases
- Treat patients
- Prescribe medicines
- Carry out medical procedures
- Promote good health

Importance

- Improves human health.
- Prevents and controls diseases.
- Saves lives.

Biology Connection

Knowledge of human anatomy, physiology, genetics, and microbiology is essential in medicine.

2. Research Scientist

A research scientist conducts investigations to discover new knowledge and solve biological problems.

Areas of Research

- Human health
- Agriculture
- Biotechnology
- Genetics
- Environmental studies

Duties

- Conduct experiments
- Collect and analyse data
- Develop new medicines
- Improve crop and animal production
- Publish scientific findings

Importance

- Advances scientific knowledge.
- Helps solve health and environmental challenges.
- Leads to technological innovations.

Biology Connection

Research scientists apply biological principles to understand living organisms and improve life.

3. Ecologist

An ecologist studies the relationships between living organisms and their environment.

Duties

- Study ecosystems
- Monitor environmental changes
- Investigate pollution effects
- Recommend conservation measures
- Conduct field surveys

Importance

- Protects ecosystems.
- Promotes sustainable use of resources.
- Helps solve environmental problems.

Biology Connection

Ecology is a branch of biology that focuses on interactions between organisms and their surroundings.

4. Conservationist

A conservationist works to protect plants, animals, and natural habitats.

Duties

- Protect endangered species
- Manage wildlife reserves
- Educate communities on conservation
- Prevent environmental degradation
- Promote sustainable resource use

Importance

- Preserves biodiversity.
- Protects natural resources.
- Maintains ecological balance.

Biology Connection

Knowledge of ecology, zoology, and botany is important for conservation work.

5. Dietician

A dietician is a health professional who advises people on proper nutrition and healthy eating.

Duties

- Plan balanced diets
- Advise patients on nutrition
- Manage nutrition-related diseases
- Promote healthy lifestyles
- Monitor dietary needs

Importance

- Prevents malnutrition.
- Improves health and well-being.
- Helps manage diseases such as diabetes and obesity.

Biology Connection

Dieticians use knowledge of human nutrition, digestion, and metabolism.

Other Biology-Related Careers

Besides the careers above, biology can lead to careers such as:

- Veterinarian
- Agricultural Officer
- Environmental Scientist
- Fisheries Officer
- Forestry Officer
- Botanist
- Zoologist
- Microbiologist
- Biotechnologist
- Laboratory Technician
- Teacher or Lecturer
- Wildlife Manager
- Public Health Officer

Importance of Studying Biology for Careers

Studying biology helps learners:

- Develop scientific skills.
- Understand living organisms.
- Solve health and environmental problems.
- Qualify for professional careers.
- Contribute to national development.

Summary Table

Career	Main Work
Medicine	Diagnosing and treating diseases
Research Scientist	Conducting scientific investigations
Ecologist	Studying organisms and their environment
Conservationist	Protecting wildlife and natural habitats
Dietician	Advising on nutrition and healthy eating

Classroom Activity

Career Research Activity

In groups, learners should:

1. Select one biology-related career.
2. Find out the duties of the professional.
3. State the biology knowledge required.
4. Present their findings to the class.

Summary

Biology provides knowledge and skills that lead to many careers. Examples include medicine, research science, ecology, conservation, and dietetics. These professions contribute to health, environmental protection, food production, and scientific development.

Evaluation Questions

Multiple Choice Questions

1. A person who studies relationships between organisms and their environment is called a:
A. Doctor
B. Ecologist
C. Dietician
D. Pharmacist

Answer: B

2. Which career mainly deals with advising people on healthy eating?
A. Conservationist
B. Ecologist
C. Dietician
D. Zoologist

Answer: C

Short Answer Questions

1. Define a career.
2. Name three biology-related careers.
3. State two duties of a research scientist.
4. Explain the work of a conservationist.
5. Give two reasons why biology is important for medicine.

Structured Question

Describe the work of:

- a) An ecologist.
- b) A dietician.

Expected Answer

- Ecologist: studies interactions between organisms and their environment, monitors ecosystems, and helps solve environmental problems.
- Dietician: advises on nutrition, plans balanced diets, and helps prevent and manage nutrition-related diseases.

Key Words

- Career
- Medicine
- Research
- Ecology
- Conservation
- Dietetics
- Nutrition
- Environment
- Biodiversity
- Scientist

Board Summary

- Biology leads to many careers.
- Major careers include medicine, research science, ecology, conservation, and dietetics.
- These careers improve health, protect the environment, and advance scientific knowledge.
- Biology knowledge is essential for success in these professions.

CHEMICALS OF LIFE

Chemicals of Life

Water

Learning objectives

By the end of the lesson, learners should be able to:

- state the constituent elements of water
- describe the chemical structure of water
- relate the properties of water to its uses in living organisms and as a habitat for living organisms

1. Introduction

Water is one of the most important chemicals of life. It is found in all living organisms and is necessary for life processes such as digestion, transport of substances, photosynthesis, excretion, and temperature control.

Water is also a habitat for many living organisms such as fish, frogs, water insects, algae, and aquatic plants.

2. Constituent elements of water

Water is made up of:

- **Hydrogen**
- **Oxygen**

Its chemical formula is **H₂O**.

This means:

- one molecule of water contains **two atoms of hydrogen**
- and **one atom of oxygen**

3. Chemical structure of water

A water molecule is formed when two hydrogen atoms bond with one oxygen atom.

Structure

- The two hydrogen atoms are joined to the oxygen atom by **covalent bonds**
- The molecule has a **bent shape**, not a straight line
- Water is a **polar molecule**, meaning it has slightly positive and slightly negative ends

Simple representation



Because of its polarity, water molecules attract each other strongly. This gives water many useful properties.

4. Properties of water and their uses in living organisms

a) Water is transparent

Water allows light to pass through it.

Use:

- It allows **aquatic plants** to receive light for **photosynthesis**
- It allows animals in water to see their surroundings

Importance in living organisms:

- Plants in water bodies can make food
- Aquatic ecosystems can support life because light reaches some depths

b) Water is a good solvent

Water dissolves many substances.

Use:

- It dissolves mineral salts in the soil so that plants can absorb them
- It dissolves food substances for transport in the body
- It dissolves wastes for excretion
- It acts as a medium for many chemical reactions in cells

Importance in living organisms:

- Nutrients are transported in blood and cell sap
- Digestive juices contain water that helps dissolve food
- Waste products such as urea are carried in solution

c) Water has a high specific heat capacity

Water needs a lot of heat to change its temperature.

Use:

- It helps to keep body temperature stable
- It prevents sudden temperature changes in aquatic habitats

Importance in living organisms:

- The body does not overheat or cool too quickly
- Aquatic organisms can survive because water temperatures change slowly

d) Water is a reagent

A reagent is a substance used in a chemical reaction.

Use:

- Water is involved in **photosynthesis**
- Water is used in **digestion**, especially in the breakdown of food
- Water is used in many metabolic reactions in cells

Importance in living organisms:

- It takes part in the formation of food in green plants
- It helps in hydrolysis, which is the breakdown of large food molecules into smaller ones

5. Relating the properties of water to its uses

Property of water	Use in living organisms
Transparent	Allows light penetration for photosynthesis in aquatic plants
Solvent	Dissolves substances for transport, digestion, and excretion
High specific heat capacity	Maintains a stable temperature in organisms and habitats
Reagent	Takes part in chemical reactions such as photosynthesis and digestion

6. Water as a habitat for living organisms

Water provides a home for many organisms. Its properties make it suitable for life.

Why water is a suitable habitat

- It is transparent, allowing light to reach plants
- It supports many dissolved substances needed by aquatic organisms
- It changes temperature slowly because of high specific heat capacity
- It contains dissolved oxygen for respiration
- It allows movement of organisms such as fish and tadpoles

Examples of aquatic organisms

- Fish
- Frogs
- Crabs
- Snails
- Water lilies
- Algae
- Plankton

7. Importance of water in living organisms

Water is needed for:

- photosynthesis
- digestion
- transport of nutrients and wastes
- excretion
- chemical reactions in cells
- temperature regulation
- support and lubrication in some body parts

8. Summary

Water is made of hydrogen and oxygen, with the formula H_2O . Its special structure gives it important properties such as transparency, solvent ability, high specific heat capacity, and its role as a reagent. These properties make water essential for life and suitable as a habitat for many organisms.

9. Evaluation questions

Short-answer questions

1. State the constituent elements of water.
2. Write the chemical formula of water.
3. Describe the structure of a water molecule.
4. State four properties of water.
5. Explain why water is a good habitat for aquatic organisms.
6. Relate the transparency of water to its role in photosynthesis.
7. Why is water called a good solvent?

Structured question

Explain how the properties of water make it useful to living organisms.

Expected points in the answer

- transparency allows light for photosynthesis
- solvent property helps transport and digestion
- high specific heat capacity helps temperature regulation
- water acts as a reagent in chemical reactions
- supports aquatic life as a habitat

10. Key vocabulary

- molecule
- hydrogen
- oxygen
- covalent bond
- polar
- solvent
- reagent
- photosynthesis
- specific heat capacity
- habitat

11. Board summary

- Water is composed of hydrogen and oxygen, H₂O.
- It has a bent, polar structure.
- Properties of water: transparent, good solvent, high specific heat capacity, reagent.
- These properties make water essential in living organisms and suitable as a habitat.

Carbohydrates, Proteins, Lipids and Nucleic Acids

Learning objectives

By the end of the lesson, learners should be able to:

- state the constituent elements of carbohydrates, proteins, lipids and nucleic acids
- identify the sub-units of these food substances
- perform tests to identify reducing sugars, non-reducing sugars, proteins and lipids

1. Introduction

Living organisms are made up of chemicals called **biomolecules** or **food substances**. These substances are important because they provide energy, build body structures, help in growth and repair, and control life processes.

The major chemical substances of life are:

- carbohydrates
- proteins
- lipids
- nucleic acids

2. Carbohydrates

(a) Constituent elements

Carbohydrates are made of:

- **Carbon (C)**
- **Hydrogen (H)**
- **Oxygen (O)**

The hydrogen and oxygen are usually in the ratio **2:1**, as in water.

(b) Sub-units of carbohydrates

The basic sub-units of carbohydrates are **simple sugars**, also called **monosaccharides**.

Examples:

- **Glucose**
- Fructose
- Galactose

Important note:

- **Glucose** is the basic unit used to form:
 - **starch**
 - **glycogen**
 - **cellulose**

How they are built:

- **Starch** = many glucose molecules joined together
- **Glycogen** = many glucose molecules joined together
- **Cellulose** = many glucose molecules joined together in a different arrangement

(c) Importance of carbohydrates

- Main source of energy
- Stored energy in plants and animals
- Structural support in plants through cellulose

3. Proteins

(a) Constituent elements

Proteins are made up of:

- **Carbon (C)**
- **Hydrogen (H)**
- **Oxygen (O)**
- **Nitrogen (N)**

Some proteins also contain:

- **Sulphur (S)**

(b) Sub-units of proteins

The basic sub-units of proteins are **amino acids**.

How proteins are formed

- Many amino acids join together to form a protein molecule
- Proteins are therefore **polymers** of amino acids

(c) Importance of proteins

- Growth and repair of tissues
- Formation of enzymes, hormones, and antibodies
- Formation of muscles, skin, and hair
- Can be used as an energy source when necessary

4. Lipids

(a) Constituent elements

Lipids are made up of:

- **Carbon (C)**
- **Hydrogen (H)**
- **Oxygen (O)**

(b) Sub-units of lipids

Lipids are made from:

- **Fatty acids**
- **Glycerol**

How lipids are formed

- One molecule of glycerol combines with fatty acids to form a fat or oil.

(c) Importance of lipids

- Store energy
- Provide insulation against cold
- Protect internal organs
- Form part of cell membranes
- Used in the production of hormones

5. Nucleic Acids

(a) Constituent elements

Nucleic acids are made up of:

- **Carbon (C)**
- **Hydrogen (H)**
- **Oxygen (O)**
- **Nitrogen (N)**
- **Phosphorus (P)**

(b) Sub-units of nucleic acids

The basic sub-units of nucleic acids are **nucleotides**.

Examples of nucleic acids:

- **DNA** (Deoxyribonucleic acid)
- **RNA** (Ribonucleic acid)

(c) Importance of nucleic acids

- Carry genetic information
- Control inheritance
- Direct protein synthesis in cells

6. Summary Table

Food substance	Constituent elements	Sub-units
Carbohydrates	Carbon, Hydrogen, Oxygen	Glucose and other simple sugars
Proteins	Carbon, Hydrogen, Oxygen, Nitrogen, sometimes Sulphur	Amino acids
Lipids	Carbon, Hydrogen, Oxygen	Fatty acids and glycerol
Nucleic acids	Carbon, Hydrogen, Oxygen, Nitrogen, Phosphorus	Nucleotides

7. Tests for food substances

A. Test for reducing sugars

Examples of reducing sugars:

- glucose
- fructose
- maltose

Reagent:

- **Benedict's solution**

Procedure:

1. Put the food solution in a test tube.
2. Add Benedict's solution.
3. Heat the mixture in a hot water bath.

Observation:

- If reducing sugar is present, the blue Benedict's solution changes to:
 - green
 - yellow
 - orange
 - brick-red precipitate

Conclusion:

A colour change from blue to orange or brick-red shows the presence of reducing sugar.

B. Test for non-reducing sugars

Example:

- sucrose

Reagents:

- dilute hydrochloric acid
- sodium hydrogencarbonate or sodium hydroxide
- Benedict's solution

Procedure:

1. Test the sample first with Benedict's solution.
2. If no colour change occurs, boil the sample with dilute hydrochloric acid.
3. Cool the solution.
4. Neutralise the acid using sodium hydrogencarbonate or sodium hydroxide.
5. Add Benedict's solution again.
6. Heat in a water bath.

Observation:

- A colour change to green, yellow, orange, or brick-red after hydrolysis shows the presence of non-reducing sugar.

Explanation:

- Non-reducing sugars are first broken down into reducing sugars by acid.
- The reducing sugars then react with Benedict's solution.

C. Test for starch

Although starch is a carbohydrate, it is often tested separately.

Reagent:

- **Iodine solution**

Procedure:

1. Add a few drops of iodine solution to the food sample.

Observation:

- **Blue-black colour** means starch is present
- **Brown/yellow-brown** means starch is absent

D. Test for proteins

Reagent:

- **Biuret reagent**
 - sodium hydroxide
 - copper(II) sulphate

Procedure:

1. Add sodium hydroxide to the sample.
2. Add a few drops of copper(II) sulphate solution.

Observation:

- **Purple/violet/lilac colour** means protein is present
- **Blue colour** means protein is absent

Note:

This is called the **Biuret test**.

E. Test for lipids

Method:

Ethanol emulsion test

Reagents:

- ethanol
- water

Procedure:

1. Add ethanol to the food sample and shake well.
2. Pour the mixture into water or add water.

Observation:

- A **white/milky emulsion** shows the presence of lipids
- If no emulsion forms, lipids are absent

Why it works:

Lipids dissolve in ethanol but not in water. When water is added, the lipid comes out as a white emulsion.

8. Results summary of food tests

Substance	Test reagent	Positive result
Reducing sugar	Benedict's solution + heat	Blue to orange/brick-red
Non-reducing sugar	Hydrolyse with acid, neutralise, then Benedict's	Blue to orange/brick-red
Starch	Iodine solution	Brown to blue-black
Protein	Biuret reagent	Blue to purple/violet
Lipid	Ethanol emulsion test	Clear to milky white emulsion

9. Precautions during food tests

- Use clean test tubes and apparatus.
- Heat Benedict's test in a water bath, not directly over a flame.
- Handle chemicals carefully.
- Use small quantities of food sample and reagents.
- Label test tubes clearly.
- Do not taste the samples.
- Wash hands after practical work.

10. Importance of these food substances in living organisms

Carbohydrates

- provide energy
- store energy
- form plant cell walls in the form of cellulose

Proteins

- growth and repair
- make enzymes and hormones
- form antibodies

Lipids

- store energy
- insulate the body
- protect organs

Nucleic acids

- carry hereditary information
- control cell activities through gene expression

11. Common misconceptions

- Not all sugars are reducing sugars.
- Starch is not a simple sugar; it is a polysaccharide made of many glucose units.
- Proteins do not give the same test as carbohydrates.
- Lipids do not dissolve in water, which is why the emulsion test works.
- Nucleic acids are not tested with the same food tests as proteins or carbohydrates.

12. Classroom practical activity

Activity: Food test station

Learners can rotate through different stations to test:

- glucose solution with Benedict's
- starch solution with iodine
- egg albumen with Biuret reagent
- cooking oil with ethanol emulsion test

They should record:

- reagent used
- colour before test
- colour after test
- conclusion

13. Revision questions

Short questions

1. State the constituent elements of carbohydrates.
2. Name the sub-units of proteins.
3. What are the sub-units of lipids?
4. State the elements found in nucleic acids.
5. What reagent is used to test for starch?
6. How do you test for reducing sugars?
7. What is the result of a positive Biuret test?
8. Explain the ethanol emulsion test for lipids.

Structured question

Describe how you would test a food sample for:

- a) reducing sugar
- b) protein
- c) lipid

Expected answer points

- reducing sugar: Benedict's solution, heat, colour change
- protein: Biuret reagent, purple/violet colour
- lipid: ethanol emulsion test, milky white emulsion

14. Key words

- carbohydrate
- protein
- lipid
- nucleic acid
- glucose
- amino acid
- fatty acid
- glycerol
- nucleotide
- reducing sugar
- non-reducing sugar
- Benedict's solution
- Biuret reagent
- emulsion
- starch

15. Board summary

- Carbohydrates: C, H, O; sub-units are simple sugars such as glucose.
- Proteins: C, H, O, N; sub-units are amino acids.
- Lipids: C, H, O; sub-units are fatty acids and glycerol.
- Nucleic acids: C, H, O, N, P; sub-units are nucleotides.
- Tests: Benedict's for reducing sugars, hydrolysis + Benedict's for non-reducing sugars, iodine for starch, Biuret for proteins, ethanol emulsion for lipids.

Cells and Cellular Activities

Sub-topic: Plant and Animal Cells

Larning objectives

By the end of the lesson, learners should be able to:

- define a cell
- identify parts of plant and animal cells
- outline the functions of cell organelles
- compare the structures of plant and animal cells

1. Introduction

All living organisms are made up of cells. Some organisms consist of only one cell, while others consist of many cells. Cells are the basic units of life and perform all life processes.

A cell is the smallest unit that can carry out life activities such as respiration, growth, nutrition, and reproduction.

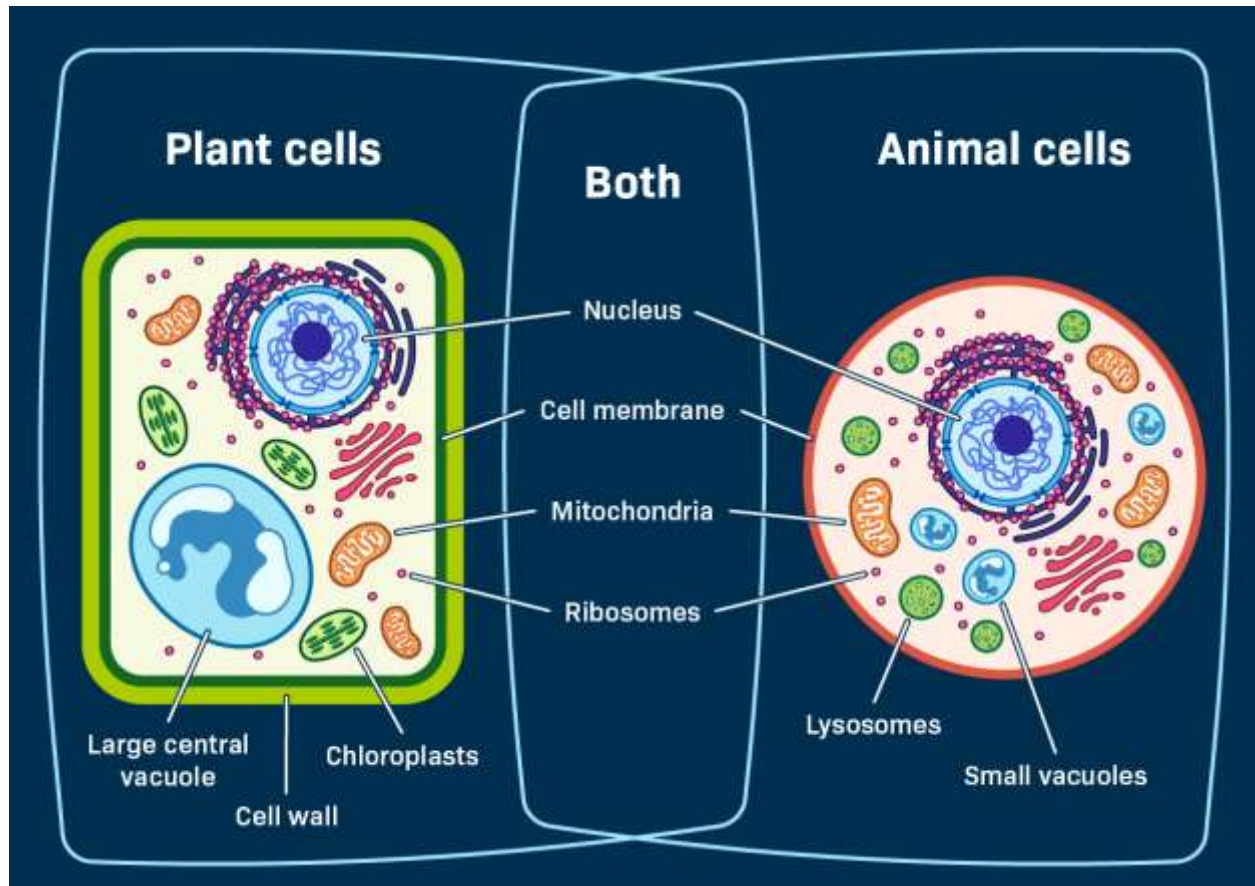
2. Definition of a cell

A **cell** is the **basic structural and functional unit of living organisms**.

This means:

- it is the smallest part of a living thing
- it performs life processes
- all organisms are made of one or more cells

3. Types of cells



There are two main types of cells at this level:

- **Plant cells**
- **Animal cells**

Both have some parts in common, but they also have differences.

4. Structure of a plant cell

A typical plant cell has the following parts:

- cell wall
- cell membrane
- cytoplasm
- nucleus
- vacuole
- mitochondria
- chloroplasts

5. Structure of an animal cell

A typical animal cell has the following parts:

- cell membrane
- cytoplasm
- nucleus
- vacuole
- mitochondria

Animal cells do **not** have:

- cell wall
- chloroplasts

6. Functions of cell organelles

a) Cell membrane

The cell membrane is a thin outer covering around the cell contents.

Function:

- controls what enters and leaves the cell
- separates the cell from its surroundings
- is selectively permeable

Key idea:

It allows only certain substances to pass through.

b) Cell wall

The cell wall is the rigid outer layer found in plant cells only.

Function:

- gives the cell shape and support
- protects the cell
- prevents the cell from bursting when it takes in too much water

Key idea:

It is made mainly of cellulose.

c) Cytoplasm

The cytoplasm is a jelly-like substance inside the cell membrane.

Function:

- holds the cell organelles in place
- is the site of many chemical reactions
- allows movement of materials within the cell

d) Nucleus

The nucleus is a round or oval structure usually found near the centre of the cell.

Function:

- controls cell activities
- contains genetic material (chromosomes/genes)
- controls cell division and inheritance

Key idea:

The nucleus is often called the “control centre” of the cell.

e) Vacuole

A vacuole is a fluid-filled sac inside the cell.

Function:

- stores cell sap, water, salts, and sugars in plant cells
- helps maintain turgidity in plant cells
- in animal cells, vacuoles are usually small or temporary

Key idea:

The large central vacuole is a feature of plant cells.

f) Mitochondria

Mitochondria are tiny structures found in the cytoplasm.

Function:

- site of respiration
- release energy needed by the cell

Key idea:

Mitochondria are called the “powerhouse” of the cell.

g) Chloroplasts

Chloroplasts are green structures found in plant cells.

Function:

- contain chlorophyll
- absorb light energy for photosynthesis
- help plants make food

Key idea:

Animal cells do not have chloroplasts because they do not photosynthesise.

7. Comparison of plant and animal cells

Feature	Plant cell	Animal cell
Cell wall	Present	Absent
Cell membrane	Present	Present
Cytoplasm	Present	Present
Nucleus	Present	Present
Vacuole	Large permanent vacuole	Small or temporary vacuoles
Mitochondria	Present	Present
Chloroplasts	Present	Absent
Shape	Usually regular/rectangular	Usually irregular/rounded

8. Similarities between plant and animal cells

Plant and animal cells both have:

- cell membrane
- cytoplasm
- nucleus
- mitochondria
- vacuoles

These similarities show that both are living cells and perform life processes.

9. Differences between plant and animal cells

Plant cell

- has a cell wall
- has chloroplasts
- has a large permanent vacuole
- usually has a regular shape

Animal cell

- has no cell wall
- has no chloroplasts
- has small or no vacuoles
- usually has an irregular shape

10. Importance of the main cell structures

Cell wall

- support
- protection
- fixed shape

Cell membrane

- controls movement of substances in and out

Cytoplasm

- site of chemical reactions

Nucleus

- controls activities
- carries hereditary information

Vacuole

- storage of cell sap
- maintains firmness in plant cells

Mitochondria

- releases energy

Chloroplast

- makes food by photosynthesis

11. Simple labelled diagram guide

Plant cell

Learners should be able to label:

- cell wall
- cell membrane
- cytoplasm
- nucleus
- large vacuole
- chloroplasts
- mitochondria

Animal cell

Learners should be able to label:

- cell membrane
- cytoplasm
- nucleus
- small vacuole
- mitochondria

12. Summary

A cell is the basic structural and functional unit of life. Plant and animal cells have some parts in common, such as the cell membrane, cytoplasm, nucleus, mitochondria, and vacuoles. Plant cells differ from animal cells because they have a cell wall and chloroplasts, and usually a large vacuole. Each organelle has a specific function that helps the cell to survive and carry out life activities.

13. Key vocabulary

- cell
- organelle
- nucleus
- cytoplasm
- cell membrane
- cell wall
- vacuole
- mitochondria
- chloroplast
- photosynthesis
- respiration
- turgidity

14. Revision questions

Short-answer questions

1. Define a cell.
2. State three parts found in both plant and animal cells.
3. What is the function of the nucleus?
4. Why are mitochondria important?
5. Name two structures found in plant cells but not in animal cells.
6. What is the function of the cell wall?
7. What is the function of chloroplasts?

Structured question

Compare plant and animal cells under the following headings:

- cell wall
- chloroplasts
- vacuole
- shape

Expected answer

- Plant cells have a cell wall, animal cells do not.
- Plant cells have chloroplasts, animal cells do not.
- Plant cells have a large permanent vacuole, animal cells have small vacuoles or none.
- Plant cells are usually regular in shape, animal cells are irregular.

15. Classroom activity

Observation activity

Learners can observe diagrams of plant and animal cells and:

- identify the organelles
- label the diagrams
- state the function of each organelle
- compare the two cell types

Practical suggestion

If a microscope is available, learners may observe:

- onion epidermal cells for plant cells
- cheek cells for animal cells

16. Board summary

- A cell is the basic structural and functional unit of life.
- Plant and animal cells both have nucleus, cytoplasm, membrane, and mitochondria.
- Plant cells have cell wall and chloroplasts.
- The nucleus controls cell activities.
- Mitochondria release energy.
- Chloroplasts make food by photosynthesis.

Cell Specialisation

Learning objectives

By the end of the lesson, learners should be able to:

- identify specialised cells
- relate cell structure to function

1. Introduction

In multicellular organisms, cells are not all the same. Different cells have different shapes, sizes, and structures because they perform different jobs.

A **specialised cell** is a cell that has developed features that help it do a particular function well.

Why cells become specialised

- to perform different tasks efficiently
- to form tissues and organs
- to allow division of labour in the body

2. Meaning of cell specialisation

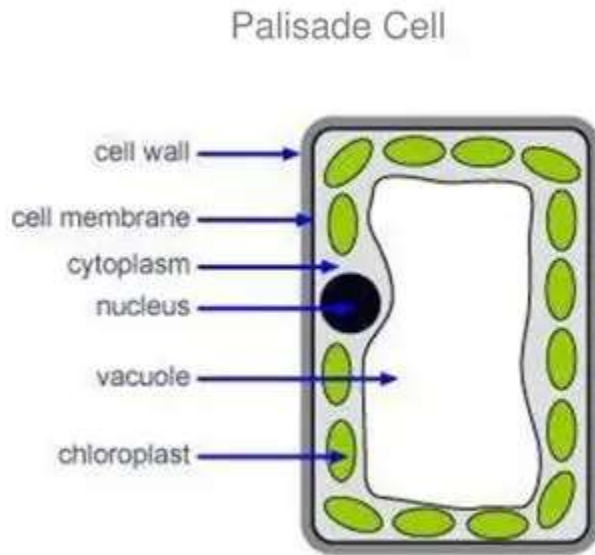
Cell specialisation is the process by which cells become adapted to carry out specific functions.

For example:

- a palisade cell is adapted for photosynthesis
- a red blood cell is adapted for oxygen transport
- a nerve cell is adapted for transmitting impulses

3. Specialised cells and their functions

a) Palisade cell



Function: photosynthesis

Adaptations

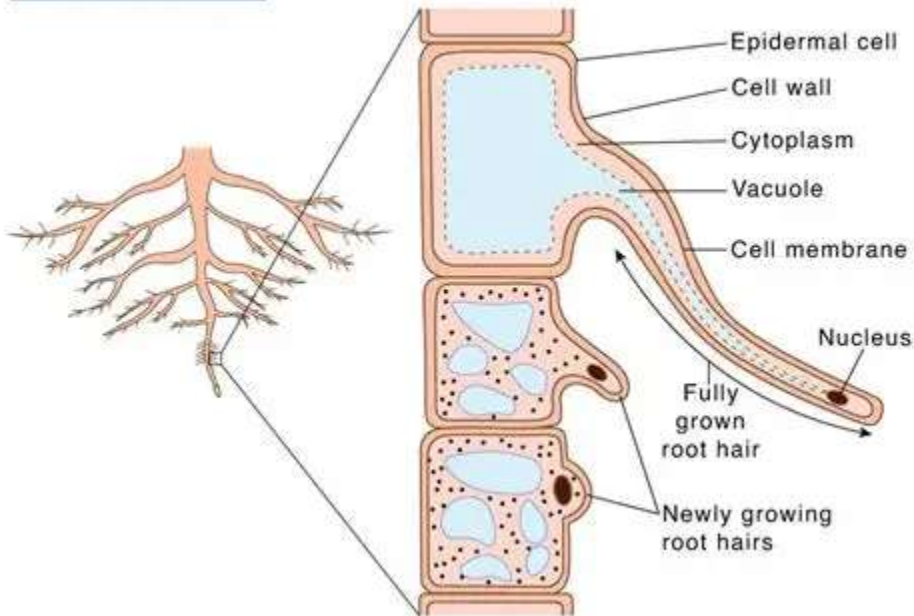
- contains many chloroplasts to trap sunlight
- has a long, upright shape to pack closely together
- thin cell wall for easier movement of gases and light penetration
- large vacuole pushes chloroplasts to the edge for better light absorption

Relation of structure to function

These features help the palisade cell make food efficiently.

b) Root hair cell

Root Hair



Function: absorption of water and mineral salts

Adaptations

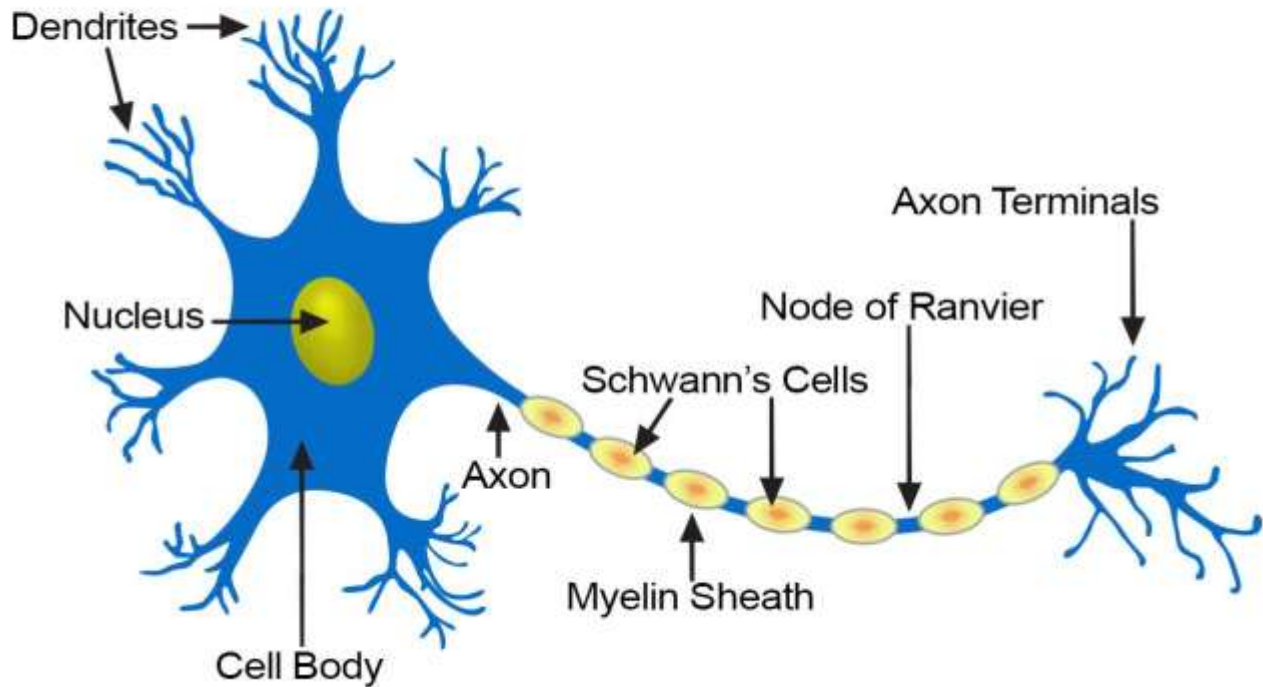
- long hair-like extension gives a large surface area
- thin cell wall allows faster absorption
- many mitochondria provide energy for active transport of mineral salts
- no chloroplasts because root cells are underground and do not photosynthesise

Relation of structure to function

The shape and thin wall help the root hair cell absorb water and minerals from the soil quickly.

c) Neurone

Structure of a Typical Neuron



Function: transmission of nerve impulses

Adaptations

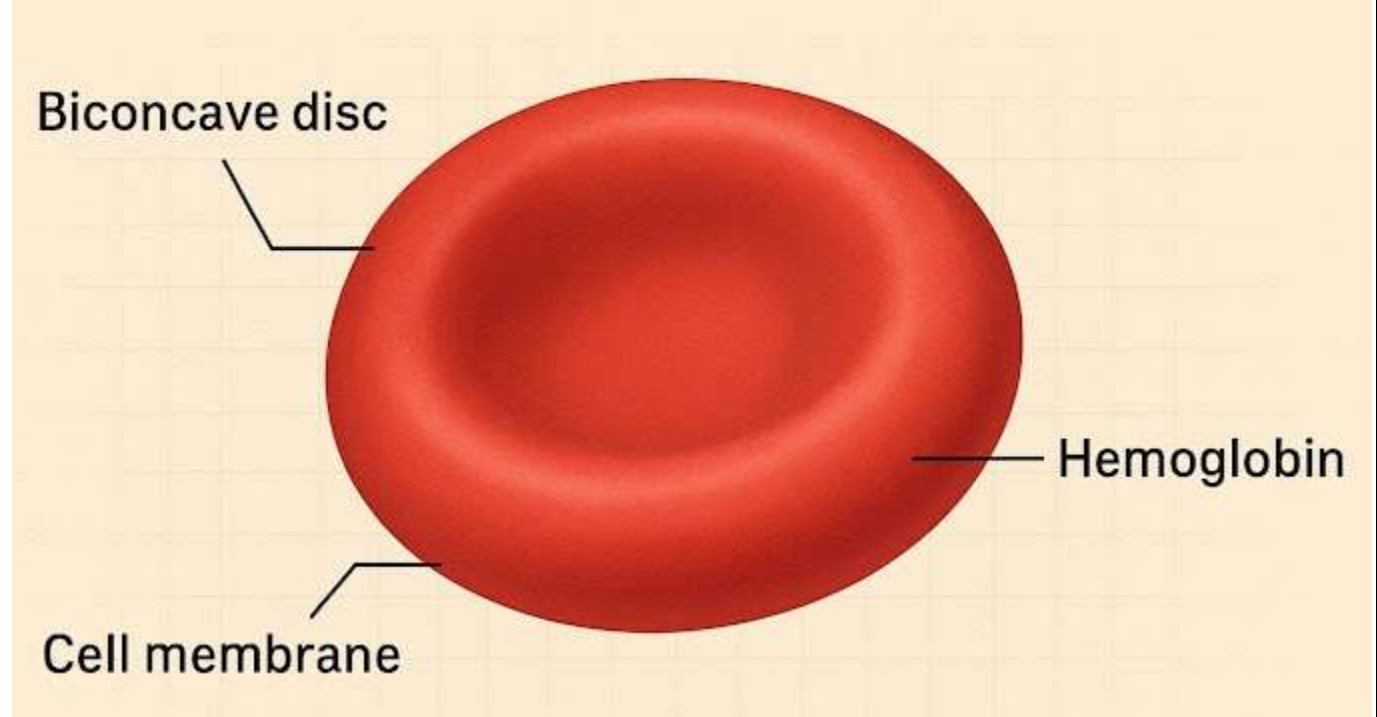
- long fibre to carry impulses over long distances
- branched ends to connect with other cells
- insulating myelin sheath on some neurones to speed up impulse transmission
- nerve endings allow communication with other neurones or muscles

Relation of structure to function

The neurone is specialised to carry messages rapidly through the body.

d) Red blood cell

STRUCTURE OF A RED BLOOD CELL



Function: transport of oxygen

Adaptations

- contains haemoglobin, which binds oxygen
- biconcave disc shape gives a large surface area
- no nucleus, so more space for haemoglobin
- flexible shape allows it to pass through narrow capillaries

Relation of structure to function

These features make the red blood cell efficient in carrying oxygen from the lungs to body tissues.

e) White blood cell

White Blood Cells (WBCs)

Granulocytes



Neutrophil

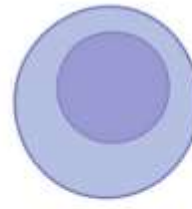


Eosinophil



Basophil

Agranulocytes



Lymphocyte



Monocyte

Function: immunity

Adaptations

- can change shape to move through tissues
- can surround and engulf germs
- contains enzymes that destroy pathogens
- some white blood cells produce antibodies

Relation of structure to function

White blood cells protect the body from disease by destroying harmful microorganisms.

f) Muscle cell

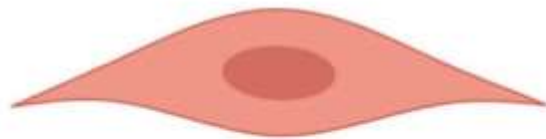
Muscle Cells



Skeletal Muscle Cell



Cardiac Muscle Cell



Smooth Muscle Cell

Function: contraction and movement

Adaptations

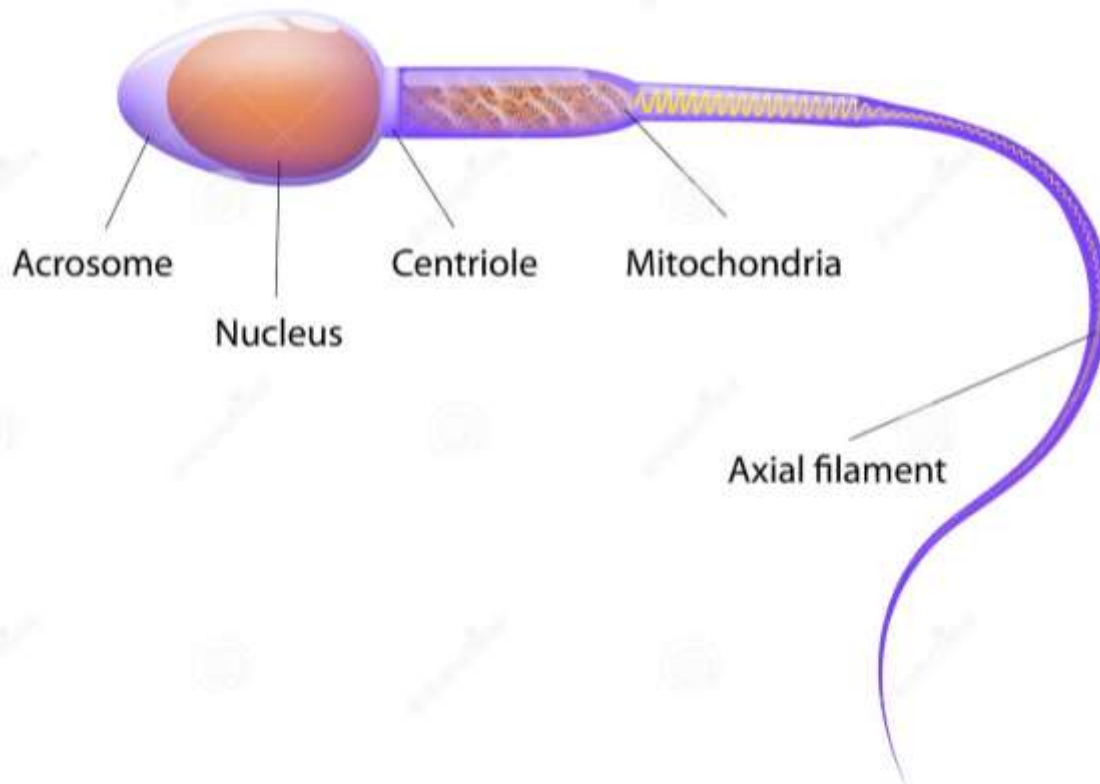
- long and fibre-like shape for contraction
- contains many mitochondria to provide energy
- contains protein filaments that slide during contraction
- cells are able to shorten and relax

Relation of structure to function

Muscle cells are adapted to contract and produce movement in the body.

g) Sperm cell

SPERM CELL



Function: fertilisation of the ovum

Adaptations

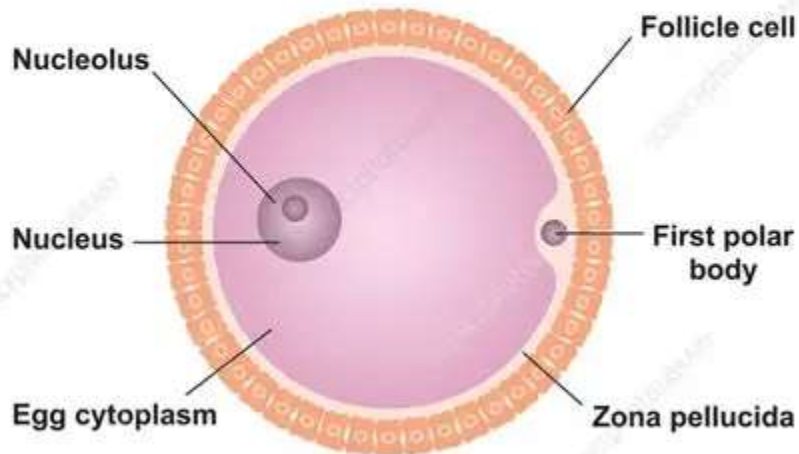
- tail or flagellum for movement
- many mitochondria for energy
- acrosome contains enzymes that help penetrate the egg
- streamlined head reduces resistance in movement
- nucleus carries genetic material

Relation of structure to function

These adaptations help the sperm swim to the ovum and fertilise it.

h) Ovum (egg cell)

Ovum Structure



Function: fertilisation and food reserve

Adaptations

- large cell with a lot of cytoplasm
- contains food reserve for early development of the embryo
- nucleus carries genetic material
- membrane changes after fertilisation to prevent entry of more sperm

Relation of structure to function

The ovum provides materials and nutrients needed for the beginning of development after fertilisation.

4. Summary table

Specialised cell	Function	Main adaptations
Palisade cell	Photosynthesis	Many chloroplasts, long shape, thin wall
Root hair cell	Absorption	Large surface area, thin wall, many mitochondria
Neurone	Transmission of impulse	Long fibre, branched ends, myelin sheath
Red blood cell	Transport of oxygen	Biconcave shape, haemoglobin, no nucleus
White blood cell	Immunity	Changes shape, engulfs germs, produces antibodies
Muscle cell	Contraction	Long fibres, many mitochondria, contractile proteins
Sperm cell	Fertilisation	Tail, acrosome, many mitochondria, streamlined
Ovum	Fertilisation and food reserve	Large size, food reserve, nucleus

5. How structure is related to function

A cell's structure is suited to its job.

Examples:

- A root hair cell has a long projection because it needs a large surface area for absorption.
- A red blood cell has no nucleus so it can carry more oxygen.
- A sperm cell has a tail so it can swim to the ovum.
- A palisade cell has many chloroplasts because it makes food by photosynthesis.

6. Importance of cell specialisation

Cell specialisation:

- increases efficiency in organisms
- allows different parts of the body to do different jobs
- helps in the formation of tissues, organs, and systems
- supports survival in complex multicellular organisms

7. Common misconceptions

- All cells are not the same in structure and function.
- Specialised cells usually perform one main task very well.
- Red blood cells in humans have no nucleus, which helps them carry more oxygen.
- Root hair cells do not have chloroplasts because they are underground.

8. Practical/classroom activity

Learners can:

- observe pictures or diagrams of specialised cells
- match each cell with its function
- list adaptations of each cell
- explain how structure helps function

Suggested group activity

Give each group one specialised cell and ask them to present:

- name of the cell
- function
- adaptations
- why the adaptations are useful

9. Revision questions

Short-answer questions

1. Define a specialised cell.
2. State the function of a palisade cell.
3. Why is a root hair cell long and thin?
4. Give two adaptations of a sperm cell.
5. Why does a red blood cell not have a nucleus?
6. State the function of a white blood cell.
7. How is a neurone adapted for its function?

Structured question

Explain how the structure of the following cells is related to their functions:

- a) red blood cell
- b) root hair cell
- c) sperm cell

10. Key vocabulary

- specialised cell
- adaptation
- photosynthesis
- absorption
- impulse
- haemoglobin
- immunity
- contraction
- fertilisation
- nucleus
- mitochondria

11. Board summary

- Specialised cells are adapted to perform specific functions.
- Structure is related to function.
- Examples: palisade cell, root hair cell, neurone, red blood cell, white blood cell, muscle cell, sperm cell, ovum.
- Each cell has features that help it do its job efficiently.

Cellular Transport

Learning objectives

By the end of the lesson, learners should be able to:

- define diffusion
- investigate factors that affect the rate of diffusion
- define osmosis
- describe active transport
- demonstrate diffusion as movement of particles in gases, liquids, and solids
- demonstrate osmosis
- explain movement against a concentration gradient

1. Introduction

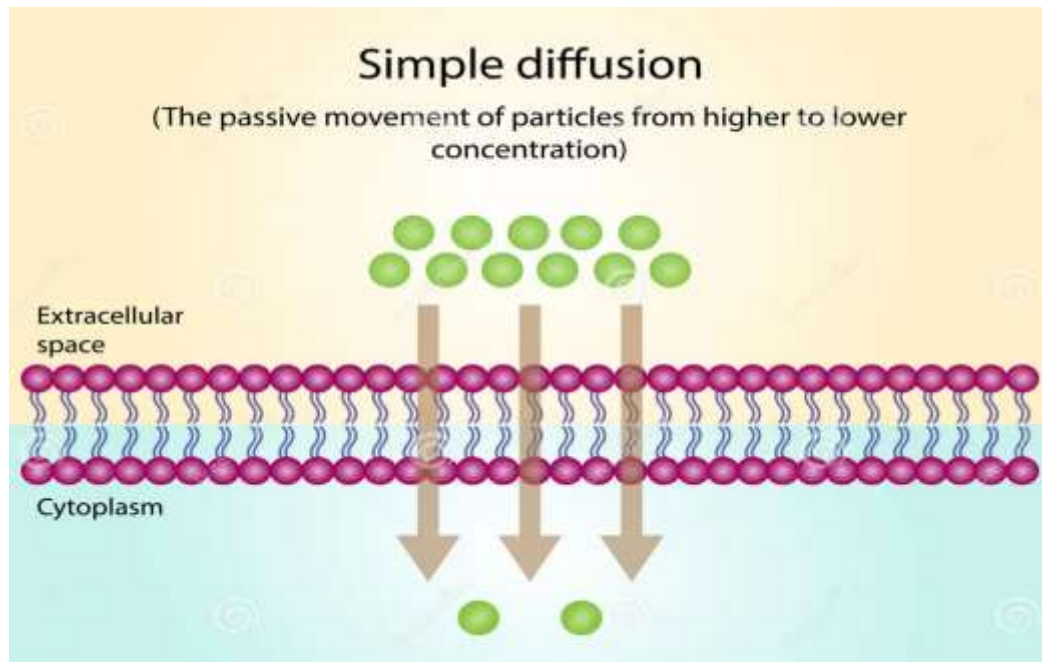
Cells need to take in useful substances such as oxygen, water, and mineral salts, and remove waste products such as carbon dioxide and urea. These materials move into and out of cells by **cellular transport**.

The main methods of cellular transport are:

- diffusion
- osmosis
- active transport

2. Diffusion

Definition



Diffusion is the movement of particles from a region of high concentration to a region of low concentration until they are evenly spread out.

Key points

- It is a **passive process**.
- It does **not require energy**.
- It happens because particles are in constant random motion.

Example

- Oxygen diffuses from the air spaces in the lungs into the blood.
- Perfume spreads through a room by diffusion.

3. Factors affecting the rate of diffusion

a) Surface area to volume ratio

A large surface area allows more particles to pass through at the same time.

Effect:

- **Large surface area** → faster diffusion
- **Small surface area** → slower diffusion

Example:

- Root hair cells have a large surface area for fast absorption of water and minerals.

b) Temperature

When temperature increases, particles gain kinetic energy and move faster.

Effect:

- **High temperature** → faster diffusion
- **Low temperature** → slower diffusion

Example:

- A smell spreads faster in a warm room than in a cold room.

c) Particle size

Small particles move more easily than large particles.

Effect:

- **Smaller particles** → faster diffusion
- **Larger particles** → slower diffusion

Example:

- Small gas molecules diffuse faster than large particles in a solution.

4. Demonstrating diffusion

Diffusion can be demonstrated in gases, liquids, and solids.

a) In gases

Example:

- Spray perfume in one corner of a room.
- After some time, the smell spreads throughout the room.

Observation:

- The perfume smell moves from a place of high concentration to low concentration.

b) In liquids

Example:

- Add a drop of potassium permanganate or ink into still water.

Observation:

- The colour gradually spreads through the water without stirring.

Explanation:

- The particles move from a concentrated area to less concentrated areas.

c) In solids

Diffusion also occurs in solids, although very slowly.

Example:

- A coloured substance may spread slowly in a gel or jelly.

Observation:

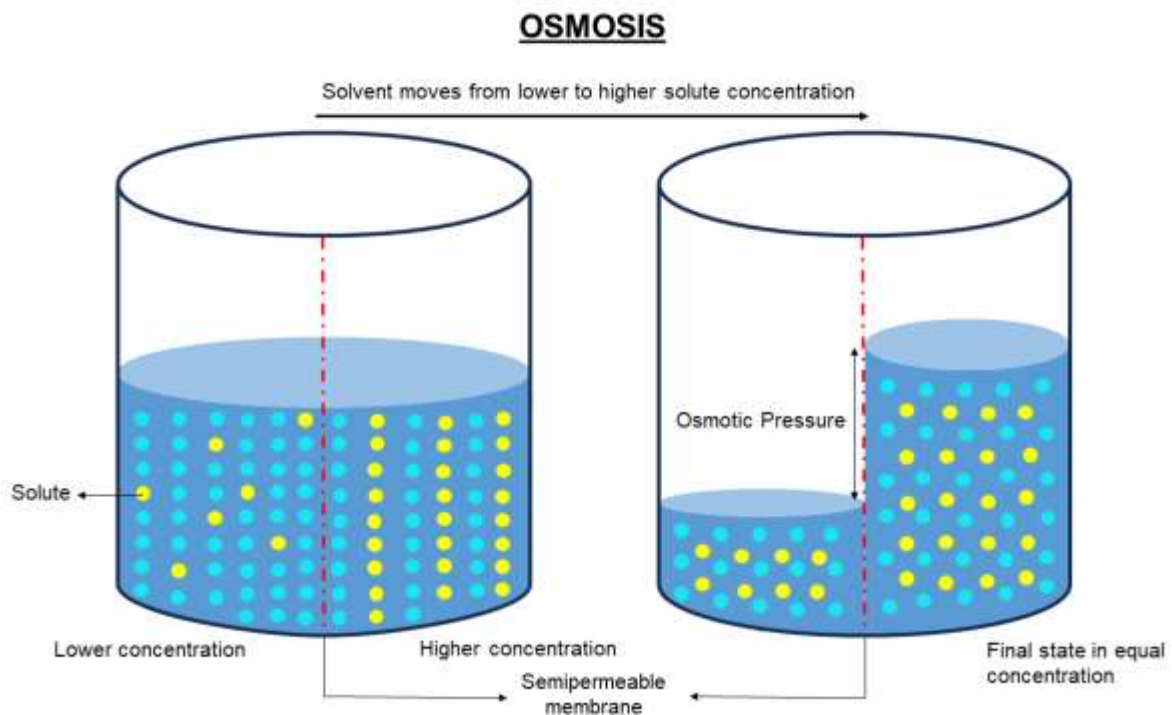
- The colour moves gradually through the solid medium.

5. Importance of diffusion in living organisms

Diffusion is important because it helps:

- oxygen enter body cells
- carbon dioxide leave cells
- food substances move into cells
- waste substances move out of cells
- water vapour move out of leaves during transpiration

7. Osmosis



Definition

Osmosis is the movement of water molecules from a region of high water concentration through a partially permeable membrane to a region of low water concentration.

Key points

- Osmosis involves **water only**
- It requires a **partially permeable membrane**
- It is a **passive process**
- No energy is needed

Partially permeable membrane

A membrane that allows some substances, such as water molecules, to pass through but not others.

7. Osmosis in living organisms

Osmosis is important in plants and animals.

In plants

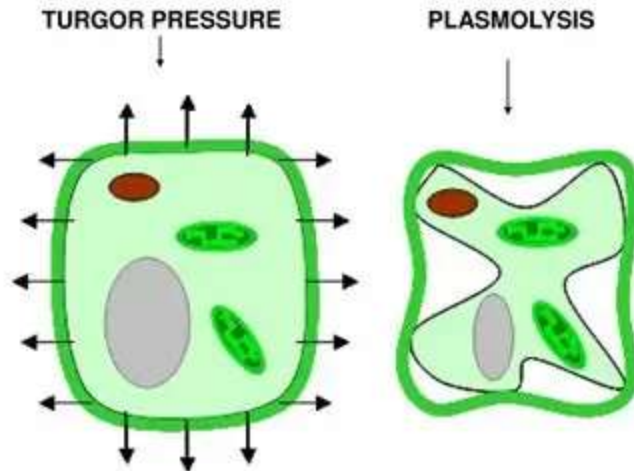
- water enters root hair cells by osmosis
- water moves from cell to cell
- helps maintain firmness in plant tissues

In animals

- water moves into and out of body cells
- helps maintain normal cell activity

8. Turgor

Turgor Pressure & Plasmolysis



Definition

Turgor is the pressure of the cell contents against the cell wall when a plant cell has absorbed water by osmosis.

Turgid cell

A plant cell that has absorbed water becomes **turgid**.

Importance of turgor

- supports non-woody plants
- keeps leaves and stems firm
- helps in opening and closing of stomata

Simple idea

When plant cells are full of water, the plant becomes firm and upright.

9. Plasmolysis

Definition

Plasmolysis is the shrinking of the cytoplasm away from the cell wall when a plant cell loses water by osmosis in a concentrated solution.

What happens

- water leaves the cell
- the vacuole becomes smaller
- the cytoplasm pulls away from the cell wall
- the cell becomes flaccid

Importance

Plasmolysis can damage plant cells if severe.

Example

- If a plant is placed in very salty water, its cells lose water and plasmolyse.

10. Demonstrating osmosis

Example experiment using a visking tubing or potato

Materials

- visking tubing or potato cup
- concentrated sugar solution
- water
- beaker
- string

Procedure

1. Fill the visking tubing with sugar solution.
2. Tie it tightly and place it in a beaker of water.
3. Leave it for some time.
4. Observe changes.

Observation

- The tubing swells as water enters it.

Explanation

- Water moves from the beaker, where it is more concentrated, into the sugar solution through the partially permeable membrane.

Another example using potato pieces

1. Place equal potato strips in water and in concentrated salt/sugar solution.
2. Leave them for some time.
3. Compare their size and firmness.

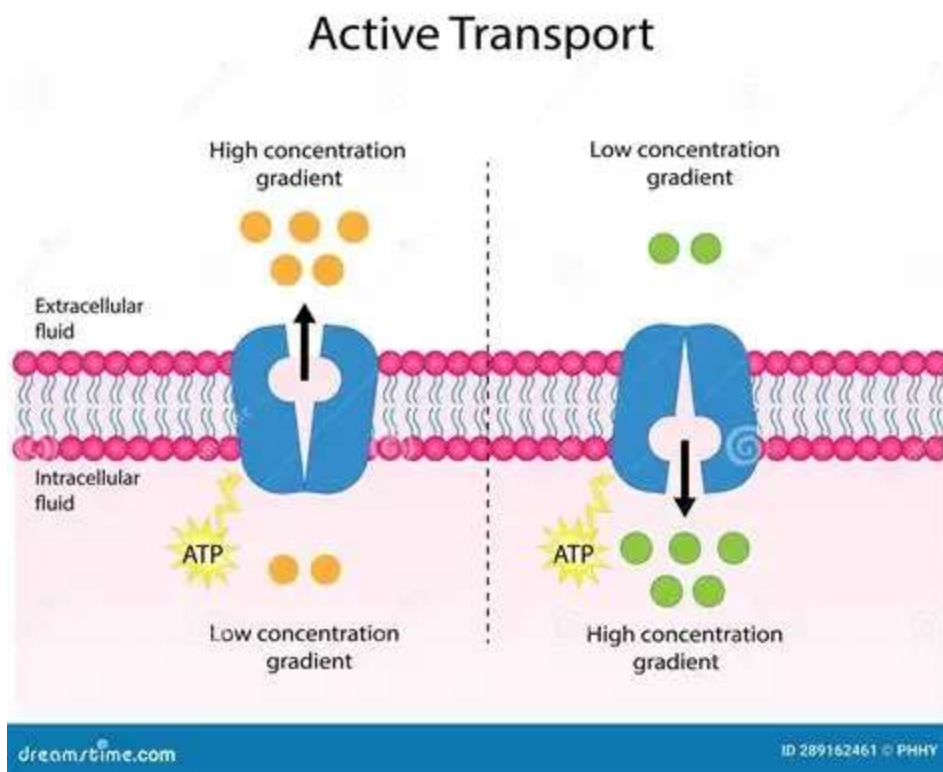
Observation

- In water, the potato becomes firm and heavy.
- In concentrated solution, the potato becomes soft and shrinks.

Explanation

- Water enters or leaves the potato cells by osmosis.

11.Active transport



Definition

Active transport is the movement of substances from a region of low concentration to a region of high concentration using energy from respiration.

Key points

- It moves **against the concentration gradient**
- It requires **energy**
- It is an **active process**

12. Movement against concentration gradient

A concentration gradient is the difference in concentration between two areas.

Diffusion

- moves **down** the concentration gradient
- from high concentration to low concentration

Active transport

- moves **against** the concentration gradient
- from low concentration to high concentration

13. Importance of active transport

Active transport is useful when cells need to take in substances even when the concentration outside is lower.

Examples

- absorption of mineral salts by root hair cells
- absorption of glucose in the small intestine
- reabsorption of useful substances in kidney tubules

14. Comparison of diffusion, osmosis and active transport

Process	What moves?	Direction	Energy needed?	Membrane required?
Diffusion	Particles	High to low concentration	No	Not always
Osmosis	Water molecules	High water concentration to low water concentration	No	Yes, partially permeable membrane
Active transport	Substances	Low to high concentration	Yes	Yes

15. Relationship between the three processes

These transport processes help organisms survive by:

- bringing in useful materials
- removing waste substances
- maintaining balance in cells
- supporting growth and metabolic activities

16. Common examples in living organisms

Diffusion

- oxygen into blood
- carbon dioxide out of blood
- perfume in air

Osmosis

- water into root hair cells
- water into plant cells
- water into animal cells

Active transport

- mineral salts into root cells
- glucose absorption in the intestine

17. Common misconceptions

- Diffusion is not the same as osmosis. Osmosis is only about water.
- Active transport needs energy, unlike diffusion and osmosis.
- Osmosis occurs through a partially permeable membrane.
- Movement against the concentration gradient is active transport, not diffusion.

18. Practical activities

Activity 1: Diffusion in liquids

Place a crystal of potassium permanganate in water and observe the spreading colour.

Activity 2: Osmosis in potato

Measure potato strips before and after placing them in water and concentrated salt solution.

Activity 3: Effect of temperature on diffusion

Compare how fast colour spreads in cold and warm water.

19. Summary

- Diffusion is the movement of particles from high to low concentration.
- Osmosis is the movement of water through a partially permeable membrane from high to low water concentration.
- Active transport is movement from low to high concentration using energy.
- These processes are essential for the survival of living organisms.

20. Key vocabulary

- diffusion
- osmosis
- active transport
- concentration gradient
- partially permeable membrane
- turgor
- plasmolysis
- passive transport
- active process
- solute
- solvent

21. Revision questions

Short-answer questions

1. Define diffusion.
2. State three factors that affect the rate of diffusion.
3. Define osmosis.
4. What is a partially permeable membrane?
5. Define active transport.
6. What is meant by movement against the concentration gradient?
7. Explain turgor.
8. What is plasmolysis?

Structured question

Compare diffusion, osmosis and active transport.

Expected points

- diffusion: particles move from high to low concentration, no energy needed
- osmosis: water moves through partially permeable membrane, no energy needed
- active transport: substances move from low to high concentration, energy needed

22. Board summary

- Diffusion: movement of particles from high to low concentration.
- Osmosis: movement of water through a partially permeable membrane from high to low water concentration.
- Active transport: movement of substances from low to high concentration using energy.
- Turgor makes plant cells firm.
- Plasmolysis happens when plant cells lose water in a concentrated solution.

Enzymes

Learning objectives

By the end of the lesson, learners should be able to:

- define an enzyme
- explain the properties of enzymes
- describe the nature of enzymes as biological catalysts
- explain enzyme action using the lock and key hypothesis
- investigate factors that affect the rate of enzyme-catalysed reactions
- describe how temperature and pH affect enzyme activity

1. Introduction

Many chemical reactions happen inside living organisms every second. These reactions must happen fast enough to sustain life. If they happened very slowly, the organism would not survive.

Enzymes make these reactions faster. They help digestion, respiration, growth, repair, and many other life processes.

2. Definition of an enzyme

An **enzyme** is a **biological catalyst**.

Meaning of biological catalyst

- **Biological** means found in living organisms
- **Catalyst** means a substance that speeds up a chemical reaction without being used up in the reaction

So, enzymes are substances made by living cells that speed up chemical reactions in the body.

3. Nature of enzymes

- Enzymes are **proteins**
- They are produced by living cells
- They work in very small amounts
- They are not used up during reactions
- Each enzyme acts on a specific substance called a **substrate**

4. Properties of enzymes

a) Enzymes are proteins

Because enzymes are proteins:

- they can be damaged by heat
- they can be affected by pH changes
- they have a specific shape

b) Enzymes are specific

Each enzyme acts on only one particular substrate or a small group of related substrates.

Example:

- **Amylase** acts on starch
- **Protease** acts on proteins
- **Lipase** acts on fats

This is called **substrate specificity**.

c) Enzymes are efficient

A small amount of enzyme can speed up a large number of reactions.

d) Enzymes are not used up

After a reaction, the enzyme remains unchanged and can be used again.

e) Enzymes work best at optimum conditions

Each enzyme works best at a certain:

- temperature
- pH

If conditions are too far from the optimum, the enzyme works less well or may stop working.

f) Enzymes can be denatured

High temperature or extreme pH can change the shape of the enzyme. When this happens, the enzyme can no longer work properly. This is called **denaturation**.

5. Substrate specificity

A **substrate** is the substance on which an enzyme acts.

Enzymes are specific because the active site of the enzyme fits only a particular substrate.

Example

- Starch is the substrate for amylase
- Protein is the substrate for protease
- Fats are the substrate for lipase

6. Mode of enzyme action

Enzymes work by combining with the substrate at a special part of the enzyme called the **active site**.

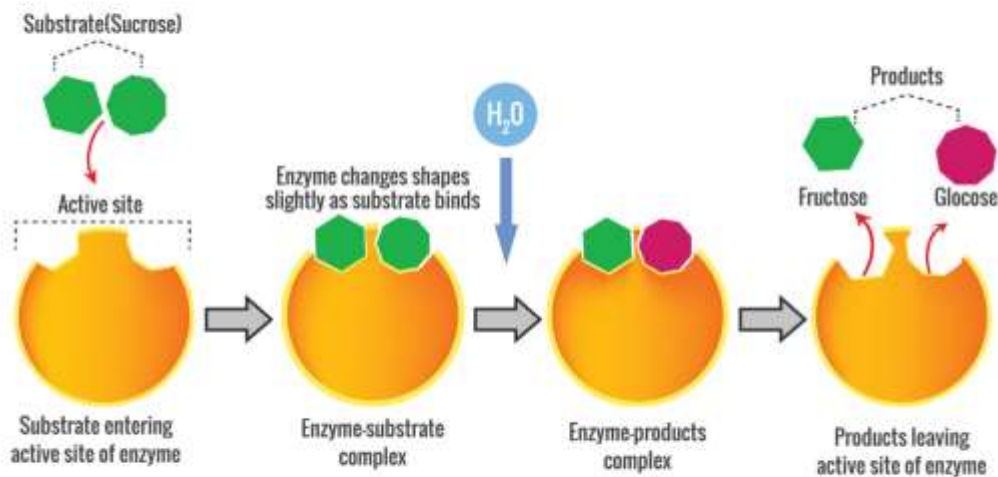
Important terms

- **Enzyme:** biological catalyst
- **Substrate:** substance acted on by the enzyme
- **Active site:** part of the enzyme where the substrate binds
- **Enzyme-substrate complex:** temporary combination formed when the substrate binds to the enzyme
- **Products:** new substances formed after the reaction

8. Lock and key hypothesis

ENZYME

Lock and Key Hypothesis



The lock and key hypothesis explains how enzymes work.

Explanation

- The enzyme has an **active site** with a specific shape
- The substrate has a shape that fits exactly into the active site
- The substrate fits into the enzyme like a **key fits into a lock**
- A temporary **enzyme-substrate complex** is formed
- The reaction occurs
- The products are released
- The enzyme remains unchanged and can be used again

Why it is called lock and key

Because only the correct substrate can fit into the enzyme's active site, just as only the correct key fits into a lock.

8. Stages of enzyme action

1. Substrate approaches the enzyme
2. Substrate fits into the active site
3. Enzyme-substrate complex is formed
4. Reaction takes place
5. Products are formed and released
6. Enzyme is reused

9. Examples of enzyme action

a) Amylase

Amylase breaks down starch into maltose, and eventually into glucose.

b) Protease

Protease breaks down proteins into amino acids.

c) Lipase

Lipase breaks down fats into fatty acids and glycerol.

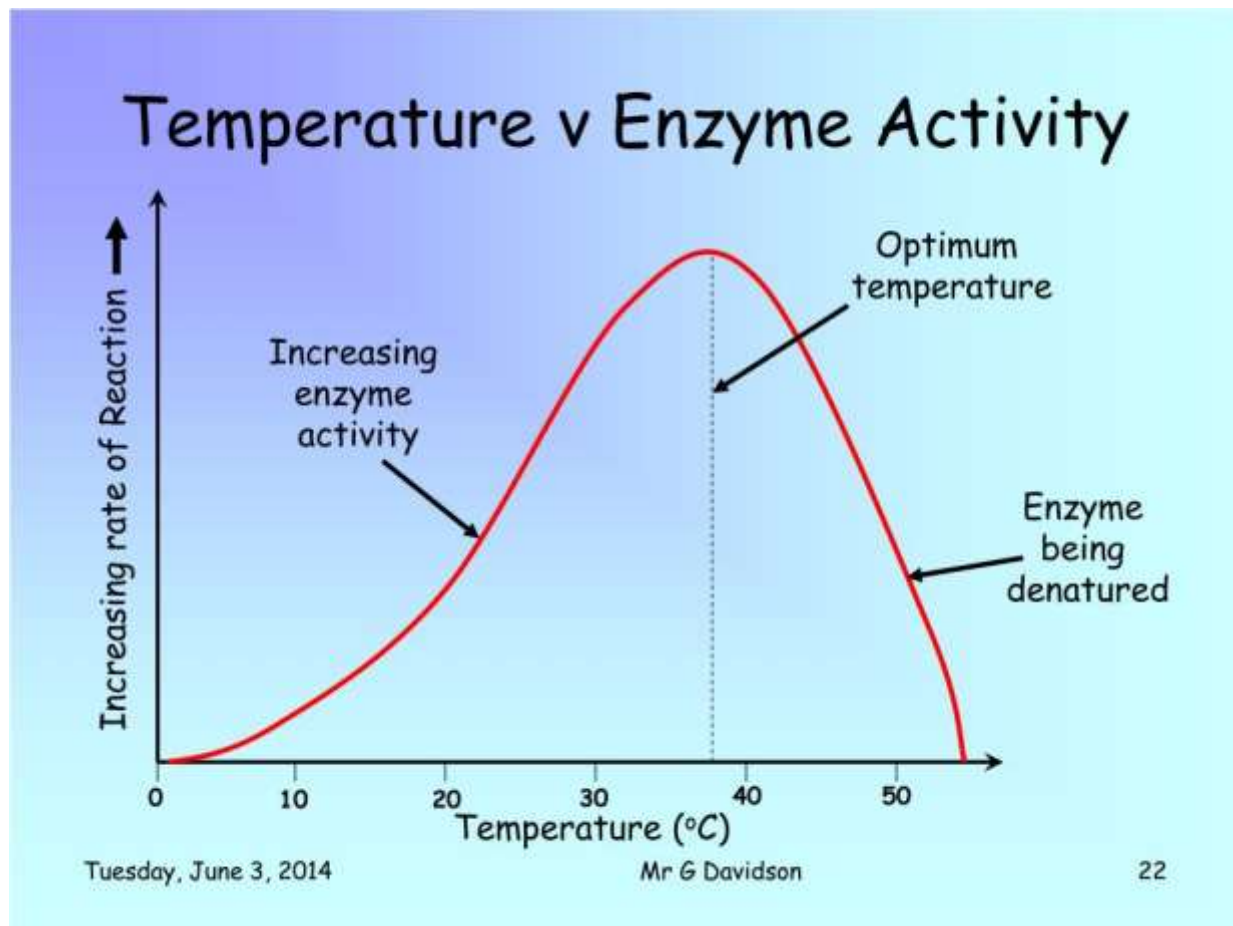
10. Factors affecting enzyme activity

The rate at which an enzyme works depends on certain conditions. The main factors are:

- temperature
- pH

Other factors, such as substrate concentration and enzyme concentration, also matter, but the objectives here focus on temperature and pH.

12. Effect of temperature on enzyme activity



Low temperature

At low temperature, enzyme and substrate particles move slowly. There are fewer successful collisions, so the reaction rate is slow.

Optimum temperature

This is the temperature at which an enzyme works fastest.

High temperature

At very high temperatures, enzymes are denatured. Their active site changes shape, so the substrate can no longer fit.

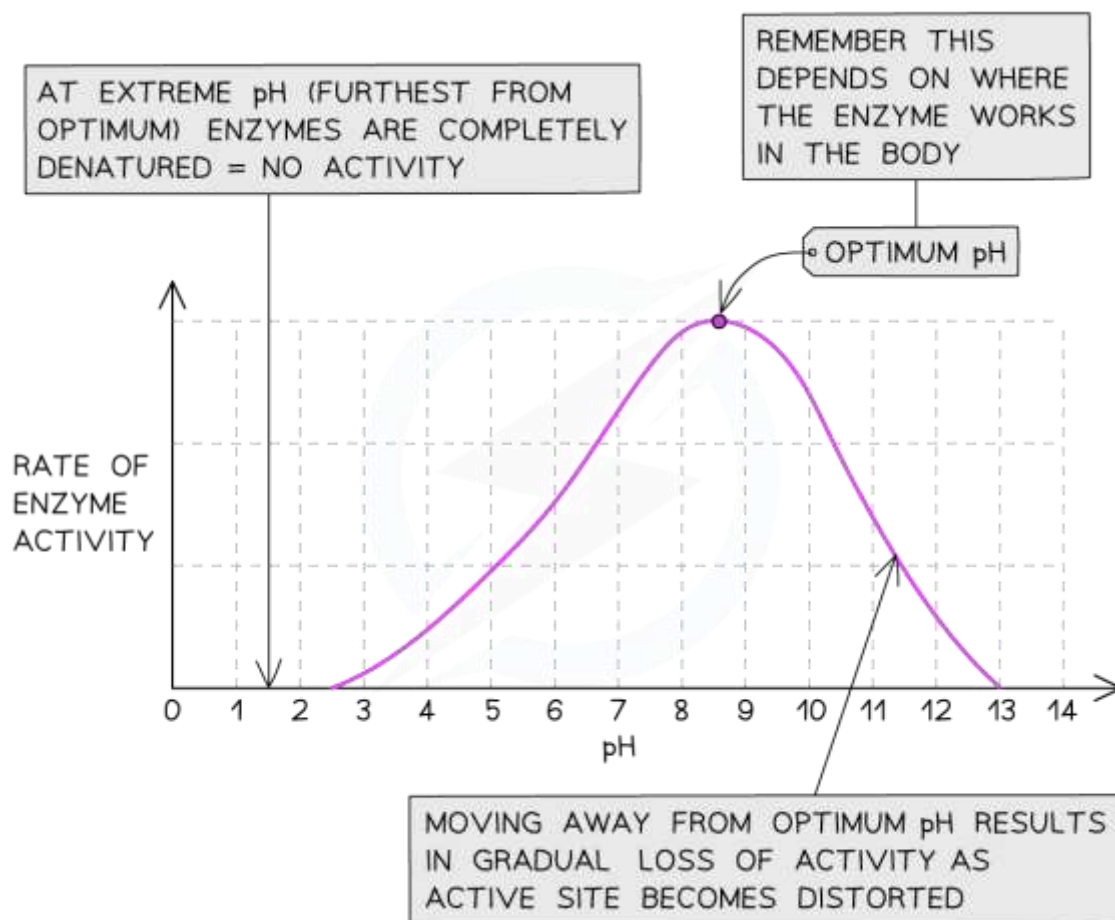
Summary of temperature effect

- Low temperature → slow reaction
- Optimum temperature → fastest reaction
- High temperature → enzyme denatures and stops working

Example

Human enzymes often work best at about 37°C

13. Effect of pH on enzyme activity



Each enzyme has an **optimum pH** at which it works best.

Low or high pH

If the pH is too acidic or too alkaline:

- the enzyme's shape may change
- the active site may no longer fit the substrate
- enzyme action slows down or stops

Examples

- **Pepsin** works best in acidic conditions in the stomach
- **Amylase** works best in neutral conditions in the mouth
- **Trypsin** works best in alkaline conditions in the small intestine

Summary of pH effect

- Optimum pH → enzyme works best
- Too acidic or too alkaline → enzyme may be denatured

13. Graphical trend of enzyme activity

When graphing enzyme activity against temperature or pH:

- the curve rises as conditions improve
- it reaches a peak at the optimum
- it falls sharply if conditions become unsuitable

This shows that enzymes work best only within a narrow range of conditions.

14. Investigating the rate of enzyme-catalysed reactions

Example practical 1: Effect of temperature on catalase

Catalase breaks down hydrogen peroxide into water and oxygen.

Materials

- potato or liver
- hydrogen peroxide
- test tubes
- water baths at different temperatures
- stopwatch

Method

1. Put equal pieces of potato or liver into separate test tubes.
2. Add the same amount of hydrogen peroxide.
3. Place the test tubes in different temperatures.
4. Measure the amount of foam or oxygen produced.

Expected result

- Faster reaction at moderate temperature
- Very slow reaction at low temperature
- Little or no reaction at very high temperature because the enzyme is denatured

Example practical 2: Effect of pH on amylase

Amylase breaks down starch.

Materials

- amylase solution
- starch solution
- buffer solutions of different pH
- iodine solution
- test tubes
- water bath

Method

1. Put starch and amylase together in different test tubes with different pH values.
2. Test samples at intervals with iodine.
3. Record the time taken for starch to disappear.

Expected result

- The enzyme works fastest at its optimum pH
- Reaction is slower at pH values far above or below the optimum

15. Precautions in enzyme experiments

- Keep all volumes the same for a fair test
- Use a control where necessary
- Measure temperature accurately
- Use fresh enzyme and substrate solutions
- Avoid contamination
- Keep pH constant when testing temperature, and keep temperature constant when testing pH

16. Importance of enzymes in living organisms

Enzymes are important because they:

- speed up digestion
- control respiration
- help in protein synthesis
- help in photosynthesis
- assist in growth and repair
- make many life processes possible at body temperature

Without enzymes, life processes would be too slow to sustain life.

17. Common misconceptions

- Enzymes are not used up in reactions.
- Enzymes are not living organisms, but they are made by living cells.
- Not all enzymes work at the same temperature or pH.
- High temperature does not make enzymes work faster forever; too much heat denatures them.
- Enzymes are specific, so one enzyme does not work on all substrates.

18. Summary

Enzymes are protein biological catalysts made by living cells. They speed up chemical reactions by combining with specific substrates at an active site. This is explained by the lock and key hypothesis. Enzyme activity is affected by temperature and pH. Low temperature slows enzyme action, optimum temperature gives the fastest reaction, and high temperature denatures enzymes. Each enzyme also has an optimum pH.

19. Key vocabulary

- enzyme
- catalyst
- biological catalyst
- protein
- substrate
- active site
- enzyme-substrate complex
- product
- specificity
- denaturation
- optimum temperature
- optimum pH
- lock and key hypothesis

20. Revision questions

Short-answer questions

1. Define an enzyme.
2. Why are enzymes called biological catalysts?
3. What is a substrate?
4. Explain the lock and key hypothesis.
5. What is denaturation?
6. State two factors that affect enzyme activity.
7. Why does high temperature affect enzyme action?
8. Give one example of an enzyme and its substrate.

Structured question

Explain how temperature affects enzyme activity.

Expected answer points

- low temperature causes slow movement of molecules
- fewer successful collisions occur
- rate increases as temperature rises to the optimum
- at very high temperatures enzymes denature
- active site changes shape and reaction stops

21. Board summary

- Enzymes are protein biological catalysts.
- They are specific to their substrates.
- Enzyme action is explained by the lock and key hypothesis.
- Enzymes are affected by temperature and pH.
- Optimum conditions give the fastest reaction.
- High temperature and extreme pH can denature enzymes.

Nutrition

Learning objectives

By the end of the lesson, learners should be able to:

- define photosynthesis
- state the word equation for photosynthesis
- investigate conditions necessary for photosynthesis
- explain the importance of photosynthesis in an ecosystem
- investigate factors affecting the rate of photosynthesis
- relate the structure of the leaf to its function in photosynthesis
- investigate gaseous exchange in plants
- describe the functions of nitrogen, phosphorus and potassium in plant growth
- describe the effects of deficiencies of nitrogen, phosphorus and potassium

1. Introduction

Plants are autotrophic organisms, meaning they make their own food. They do this mainly by photosynthesis. Photosynthesis is one of the most important processes on Earth because it provides food and oxygen for most living organisms.

2. Definition of photosynthesis

Photosynthesis is the process by which green plants use light energy and chlorophyll to make food from carbon dioxide and water, releasing oxygen.

This process takes place mainly in the leaves, especially in cells containing chloroplasts.

3. Word equation for photosynthesis

carbon dioxide + water → glucose + oxygen
light energy and chlorophyll are needed

4. Conditions necessary for photosynthesis

Photosynthesis needs the following conditions:

a) Light

- Light provides the energy needed for photosynthesis.
- Without light, the process cannot occur.

b) Chlorophyll

- Chlorophyll is the green pigment found in chloroplasts.
- It traps light energy.

c) Carbon dioxide

- Carbon dioxide is taken from the air through stomata in the leaf.
- It is one of the raw materials for photosynthesis.

d) Water

- Water is absorbed from the soil by roots and transported to the leaves.
- It is also a raw material for photosynthesis.

e) Suitable temperature

- Photosynthesis is controlled by enzymes.
- A suitable temperature is needed for enzymes to work properly.

5. Investigating conditions necessary for photosynthesis

Experiment 1: Testing for starch in a leaf

A leaf that has photosynthesised stores some of the food as starch.

Procedure

1. Keep a plant in the dark for 24–48 hours to remove starch from the leaves.
This is called **destarching**.
2. Expose the plant to sunlight.
3. Remove a leaf and boil it in water.
4. Heat it in alcohol using a water bath to remove chlorophyll.
5. Rinse the leaf in warm water.
6. Add iodine solution.

Observation

- The leaf turns **blue-black** if starch is present.

Conclusion

- Starch was made in the leaf, showing that photosynthesis took place.

Experiment 2: Variegated leaf experiment

A variegated leaf has green and non-green parts.

Procedure

1. Expose a destarched variegated plant to sunlight.
2. Test the leaf for starch after some time.

Observation

- Only the green parts turn blue-black with iodine.

Conclusion

- Chlorophyll is necessary for photosynthesis.

Experiment 3: Carbon dioxide is necessary

Procedure

1. Place a destarched plant in a sealed bell jar.
2. Put potassium hydroxide or soda lime inside the jar to absorb carbon dioxide.
3. Leave another plant as a control.
4. Expose both to light.
5. Test the leaves for starch.

Observation

- The plant without carbon dioxide does not form starch.

Conclusion

- Carbon dioxide is necessary for photosynthesis.

Experiment 4: Light is necessary

Procedure

1. Cover part of a leaf with opaque black paper.
2. Expose the plant to sunlight.
3. Test the leaf for starch.

Observation

- The uncovered part turns blue-black.
- The covered part remains brown or yellow.

Conclusion

- Light is necessary for photosynthesis.

6. Fate of the end products of photosynthesis

The main product of photosynthesis is **glucose**. It can be used in several ways:

a) Used in respiration

- Glucose is broken down during respiration to release energy.

b) Stored as starch

- Excess glucose is converted to starch for storage.

c) Used to make cellulose

- Glucose is used to build cellulose for cell walls.

d) Used to make fats and oils

- In some plants, glucose is converted into oils or fats.

e) Used to make proteins

- Glucose combines with nitrates absorbed from the soil to form amino acids and then proteins.

7. Importance of photosynthesis in an ecosystem

Photosynthesis is important because it:

a) Produces food

- It is the main source of food for plants and animals.
- Animals depend directly or indirectly on plants for food.

b) Produces oxygen

- Oxygen released during photosynthesis is used in respiration by living organisms.

c) Maintains the balance of gases

- It removes carbon dioxide from the atmosphere and adds oxygen.

d) Supports food chains and food webs

- Plants are the producers at the beginning of most food chains.

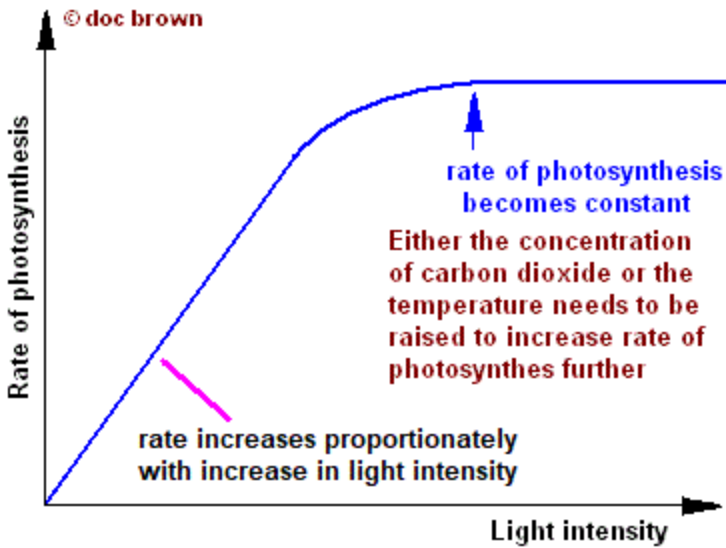
e) Provides raw materials

- Products of photosynthesis are used in industry and agriculture.

f) Helps store energy

- Light energy is converted into chemical energy in glucose.

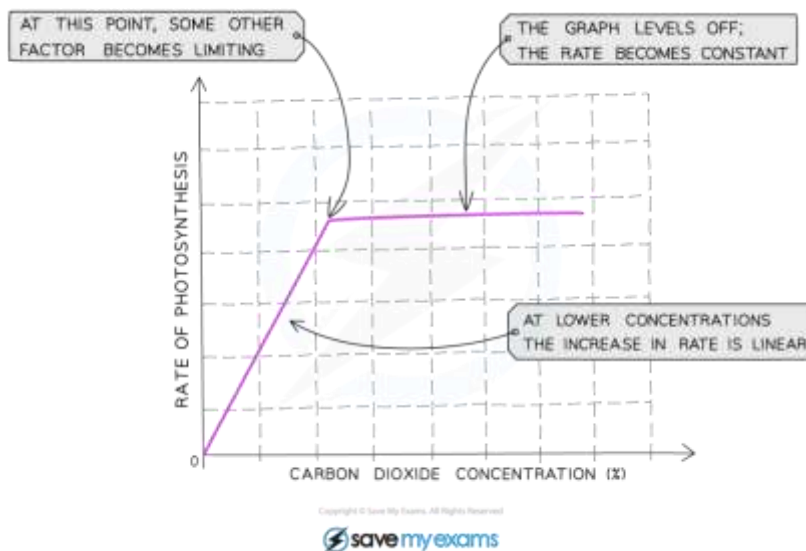
7. Factors affecting the rate of photosynthesis



a) Light intensity

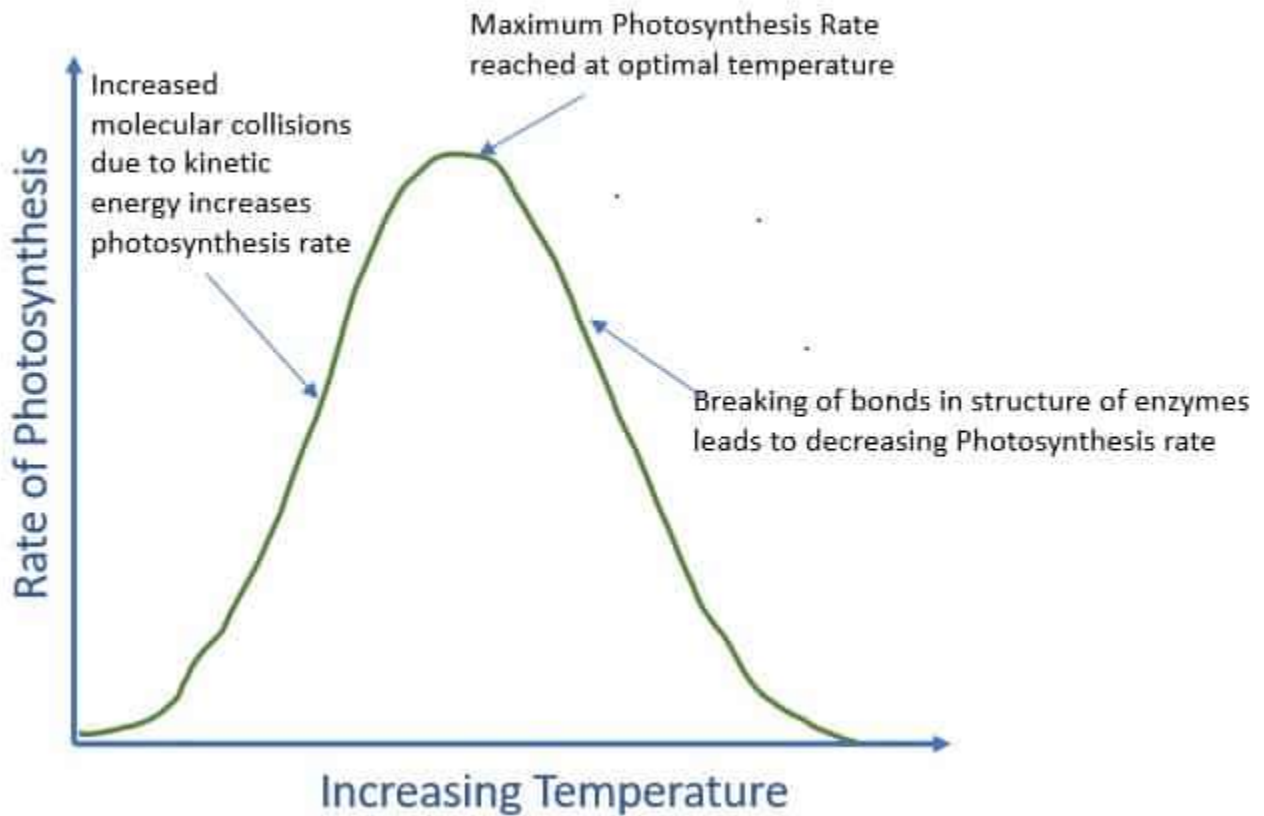
- Increasing light intensity increases the rate of photosynthesis up to a certain point.
- At low light intensity, photosynthesis is slow.
- At very high light intensity, the rate levels off because other factors become limiting.

b) Carbon dioxide concentration



- More carbon dioxide generally increases the rate of photosynthesis.
- When carbon dioxide becomes sufficient, another factor limits the rate.

c) Temperature



- Photosynthesis is controlled by enzymes.
- Low temperature slows enzyme activity.
- Very high temperature may damage enzymes and reduce photosynthesis.

Other factors sometimes considered

- chlorophyll content
- water availability
- mineral nutrients
- leaf area

9. Limiting factors

A limiting factor is the factor that is in shortest supply and therefore slows down the rate of photosynthesis.

For example:

- if light is low, increasing carbon dioxide will not help much
- if temperature is too low, photosynthesis remains slow even if light is strong

10. Relating the structure of the leaf to its function in photosynthesis

The leaf is specially adapted for photosynthesis.

a) Broad and flat shape

- Provides a large surface area for absorbing light.

b) Thin leaf

- Allows gases to diffuse quickly.
- Light penetrates easily to cells inside.

c) Many chloroplasts in palisade cells

- Chloroplasts trap light energy.

d) Transparent upper epidermis

- Allows light to reach the photosynthetic cells.

e) Stomata

- Allow carbon dioxide to enter and oxygen to leave.

f) Spongy mesophyll with air spaces

- Air spaces allow rapid diffusion of gases.

g) Veins containing xylem and phloem

- Xylem carries water to the leaf.
- Phloem carries food away from the leaf.

h) Waxy cuticle

- Reduces water loss while still allowing light through.

11. Gaseous exchange in leaves

Gaseous exchange is the movement of gases in and out of the leaf.

How it happens

- Carbon dioxide enters the leaf through stomata.
- Oxygen produced during photosynthesis leaves the leaf through stomata.
- Water vapour also escapes during transpiration.

Role of stomata

- Stomata are tiny pores in the leaf epidermis.
- They open and close by the action of guard cells.
- They regulate gas exchange and water loss.

Importance

- Carbon dioxide is needed for photosynthesis.
- Oxygen is needed for respiration.
- Gaseous exchange keeps the plant alive and functioning

12. Functions of mineral nutrients in plant growth

Plants need mineral salts from the soil for healthy growth. The main nutrients in this topic are nitrogen, phosphorus and potassium.

a) Nitrogen (N)

Function

- needed for protein synthesis
- needed for chlorophyll formation
- supports leafy growth

Importance

Nitrogen helps plants make amino acids and proteins, which are necessary for growth and repair.

b) Phosphorus (P)

Function

- needed for synthesis of ATP
- needed for nucleic acids
- supports root development and energy transfer

Importance

Phosphorus is important in energy transfer and genetic material formation.

c) Potassium (K)

Function

- helps maintain ion balance
- helps maintain osmotic balance
- helps normal enzyme activity
- supports flower and fruit development

Importance

Potassium helps regulate many processes in the plant, including water balance and healthy growth.

14. Deficiency symptoms

15.

a) Nitrogen deficiency

Symptoms

- stunted growth
- chlorosis, especially in older leaves

Explanation

Without enough nitrogen, the plant cannot make enough protein or chlorophyll.

b) Phosphorus deficiency

Symptoms

- stunted root growth
- purplish leaf colouring
- poor growth

Explanation

Lack of phosphorus affects ATP production and nucleic acid synthesis.

c) Potassium deficiency

Symptoms

- yellow and brown leaf margins
- premature death of leaves
- poor flowering and fruiting

Explanation

Without enough potassium, the plant cannot regulate ions and water balance properly.

14. Summary table of mineral nutrients

Nutrient	Main function	Deficiency symptoms
Nitrogen	Protein and chlorophyll formation	Stunted growth, chlorosis
Phosphorus	ATP and nucleic acid synthesis	Stunted root growth, purplish leaves
Potassium	Ion and osmotic balance	Yellow/brown leaf margins, poor flowering, premature death

15. Importance of mineral nutrients to plants

Mineral nutrients help plants to:

- grow well
- make proteins
- photosynthesise efficiently
- form strong roots
- flower and fruit properly
- resist disease

16. Common misconceptions

- Photosynthesis does not occur in all plant cells, only in cells with chloroplasts.
- Oxygen is a product of photosynthesis, not a raw material.
- Plants need carbon dioxide, not oxygen, for photosynthesis.
- A plant can have chlorophyll and still not photosynthesise if other conditions are missing.
- Mineral deficiencies do not always kill plants immediately, but they reduce growth and yield.

17. Practical activities

Activity 1: Starch test in a leaf

Use iodine solution to test for starch after exposing a plant to light.

Activity 2: Leaf adaptation observation

Observe a leaf and identify features that help photosynthesis.

Activity 3: Stomata observation

Use a microscope to observe leaf epidermis and stomata.

Activity 4: Mineral deficiency observation

Compare healthy plants with plants showing deficiency symptoms.

19. Revision questions

Short-answer questions

1. Define photosynthesis.
2. Write the word equation for photosynthesis.
3. Name three conditions necessary for photosynthesis.
4. Why is chlorophyll important in photosynthesis?
5. State two uses of glucose in plants.
6. How are leaves adapted for photosynthesis?
7. What is the function of stomata?
8. State two symptoms of nitrogen deficiency.
9. Give two symptoms of phosphorus deficiency.
10. What are the symptoms of potassium deficiency?

Structured question

Explain how the structure of a leaf is adapted for photosynthesis.

Expected points

- broad surface area
- thin leaf
- transparent epidermis
- palisade cells with many chloroplasts
- stomata for gas exchange
- air spaces for diffusion
- xylem brings water
- phloem carries food away

20. Board summary

- Photosynthesis is the process by which green plants make food using light, carbon dioxide, water and chlorophyll.
- Word equation: carbon dioxide + water → glucose + oxygen.
- Conditions needed: light, chlorophyll, carbon dioxide, water, suitable temperature.
- Photosynthesis is important because it provides food and oxygen and maintains gas balance.
- Light intensity, carbon dioxide concentration and temperature affect the rate of photosynthesis.
- Leaves are adapted for photosynthesis and gaseous exchange.
- Nitrogen, phosphorus and potassium are essential mineral nutrients.
- Deficiency symptoms help us identify nutrient shortages in plants.

Productivity

Learning objectives

By the end of the lesson, learners should be able to:

- define biomass
- define productivity
- explain factors affecting productivity
- define a pest
- identify types of plant pests and diseases
- explain how pests and diseases affect productivity
- describe methods of controlling pests and diseases
- outline the advantages and disadvantages of each method

1. Introduction

Productivity in plants is important because it shows how much useful plant material is produced over a period of time. Good productivity means better harvests, more food, and more income for farmers.

Plants may grow well only when conditions are favourable. When pests and diseases attack crops, productivity falls.

2. Biomass

Definition

Biomass is the total organic matter or organic content of an organism or group of organisms.

In plants, biomass refers to the mass of living material such as:

- stems
- leaves
- roots
- fruits
- seeds

Key idea

Biomass is the **dry mass** or organic material produced by living things. It is a measure of growth.

Example

A maize plant that has many leaves, a tall stem, and large cobs has a greater biomass than a small weak maize plant.

3. Productivity

Definition

Productivity is the increase in biomass over time.

It shows how fast plants are producing new living material.

Simple meaning

If a plant or crop gains mass quickly, it is said to have high productivity.

Example

A field of maize that grows quickly and produces many grains has high productivity.

4. Relationship between biomass and productivity

- **Biomass** is the amount of organic material present.
- **Productivity** is the rate at which biomass increases.

So:

- biomass is the product
- productivity is the process or rate of production

5. Factors affecting productivity

Several factors affect how much biomass plants produce.

a) Light

Light is needed for photosynthesis.

Effect on productivity

- More light usually increases photosynthesis.
- More photosynthesis means more food made.
- More food leads to more growth and higher productivity.

If light is too low

- photosynthesis slows down
- growth becomes poor
- biomass production decreases

b) Mineral salts

Plants need mineral nutrients from the soil.

Important mineral salts

- nitrogen
- phosphorus
- potassium

Effect on productivity

- Nutrients help plants make proteins, chlorophyll, ATP, and other important substances.
- Healthy nutrition leads to better growth and higher yields.

If mineral salts are lacking

- plants become weak
- growth is stunted
- productivity falls

c) Temperature

Temperature affects enzyme activity and photosynthesis.

Effect on productivity

- Suitable temperature increases enzyme action and growth.
- Very low temperature slows metabolism.
- Very high temperature may damage enzymes.

Result

Productivity is highest at an optimum temperature.

d) Water availability

Water is needed for:

- photosynthesis
- transport of mineral salts
- turgor
- enzyme reactions

Effect on productivity

- enough water helps plants grow well
- water shortage causes wilting and poor growth

If water is insufficient

- stomata close
- photosynthesis decreases
- productivity falls

e) Pests and diseases

Pests and diseases reduce healthy plant growth.

Effect on productivity

- they damage leaves, stems, roots, flowers, fruits, and seeds
- they reduce photosynthesis and nutrient absorption
- they lower yields

6. Definition of a pest

A pest is an organism that reduces productivity by damaging crops or interfering with plant growth.

Examples of pests

- insects
- nematodes
- rodents
- birds
- weeds can also be considered harmful to crops in some contexts

7. Types of plant pests and diseases

A. Plant pests

Damage	Pest
Deformed leaves, sucking damage 	Aphids 
Discolored leaves, sucking damage 	Thrips and mites 
Chewed or skeletonized leaves 	Beetles, caterpillars, and sawflies 
Leaf galls (abnormal plant growths) 	Cynipid wasps, certain aphids, psyllids, and mites 
Leaf mines (white patterns on leaves) 	Beetle, fly, or moth larvae 
Folded leaves 	Caterpillars, tree crickets, and spiders 
Rolled leaves 	Certain mites or some caterpillars 
Chewed leaves, slime trails 	Slugs and snails 

Plant pests are organisms that feed on or damage crops.

1. Tissue-eating pests

These pests eat leaves, stems, roots, flowers, or fruits.

Examples

- caterpillars
- beetles
- grasshoppers
- armyworms
- termites
-

Damage caused

- holes in leaves
- destruction of stems or roots
- reduced photosynthesis
- poor growth and yield loss

2. Sap-sucking pests

These pests suck sap from plants.

Examples

- aphids
- whiteflies
- scale insects
- mealybugs
- leafhoppers

Damage caused

- wilting
- yellowing of leaves
- stunted growth
- spread of diseases
- reduced productivity

B. Plant diseases

Plant diseases are caused by pathogens such as bacteria and fungi.

Examples

- bacterial wilt
- fungal rust
- blight
- mosaic disease
- rot diseases

8. Examples of plant diseases

a) Bacterial wilt



This is a disease caused by bacteria that block water movement in the plant.

Symptoms

- wilting of leaves and stems
- plant may collapse
- yellowing
- death of the plant

Effect on productivity

- plants cannot transport water properly
- crop yield is reduced

b) Fungal rust

Rust is a fungal disease that appears as rust-coloured spots or pustules on leaves and stems.



Symptoms

- orange, brown, or reddish spots
- weak leaves
- leaf damage
- reduced photosynthesis

Effect on productivity

- less food is made
- seeds and fruits may develop poorly
- yield decreases

9. How pests and diseases affect productivity

Pests and diseases reduce productivity in several ways:

- damage leaves, reducing photosynthesis
- damage roots, reducing water and mineral uptake
- damage stems, weakening support and transport
- destroy flowers and fruits, reducing yield
- spread infections from plant to plant
- cause premature death of plants
- reduce the quality of crops

Result

The crop produces less biomass and fewer marketable products.

10. Methods of controlling pests and diseases

Plant pests and diseases can be controlled by different methods. These include:

- management methods
- cultural methods
- chemical methods
- biological methods

11. Management methods

Management methods involve planning and monitoring to reduce pest and disease damage.

Examples

- regular field inspection
- early detection of pests and disease symptoms
- use of clean planting materials
- quarantining infected plants
- removing and destroying infected parts

Advantages

- helps detect problems early
- reduces spread of disease
- may be cheap compared to chemicals
- supports better farm management

Disadvantages

- requires constant supervision
- may need trained personnel
- not always enough on its own

12. Cultural methods

Cultural methods are farming practices used to reduce pests and diseases.

Examples

- crop rotation
- proper spacing
- proper tilling and weeding
- field sanitation
- removal of crop residues
- use of resistant varieties
- early planting
- intercropping
- proper irrigation and drainage

Advantages

- cheap and environmentally friendly
- reduces need for chemicals
- improves soil health
- can prevent many pest and disease outbreaks

Disadvantages

- may take time to show results
- not always fully effective in severe outbreaks
- requires knowledge and planning

13. Chemical methods

Chemical methods involve the use of chemicals to kill pests or control diseases.

Examples

- insecticides
- fungicides
- pesticides
- herbicides, in some crop protection contexts

Advantages

- fast acting
- effective against many pests
- useful during severe infestations

Disadvantages

- may pollute the environment
- may kill useful organisms
- can be dangerous to humans and animals
- pests may become resistant
- expensive for some farmers

Safety note

Chemicals must be used carefully and according to instructions.

14. Biological methods

Biological control uses living organisms to control pests.

Examples

- introducing natural enemies of pests
- using predators such as ladybirds to eat aphids
- using parasites or pathogens that attack pests

Advantages

- environmentally friendly
- reduces chemical use
- can be long-lasting
- does not usually leave harmful residues

Disadvantages

- works slowly
- may not eliminate pests completely
- requires careful study before introduction
- may affect other organisms if not managed properly

15. Comparison of control methods

Method	Advantages	Disadvantages
Management	Early detection, useful in planning	Needs regular monitoring
Cultural	Cheap, safe, eco-friendly	May not be enough alone
Chemical	Fast and effective	Pollution, resistance, health risks
Biological	Safe and natural	Slow and sometimes less predictable

16. Control of pests and diseases in maize, cotton and tobacco

A. Maize

Common problems in maize may include:

- stem borers
- armyworms
- aphids
- fungal diseases such as rust or blight

Control methods

- crop rotation
- early planting
- use of resistant varieties
- destruction of infected plants
- proper field sanitation
- spraying when necessary

B. Cotton

Common pests and diseases in cotton may include:

- bollworms
- aphids
- whiteflies
- bacterial and fungal diseases

Control methods

- crop rotation
- proper spacing
- use of clean seed
- field scouting
- biological control
- use of chemicals when infestation is severe

C. Tobacco

Common pests and diseases in tobacco may include:

- aphids
- caterpillars
- fungal diseases
- bacterial wilt in some areas

Control methods

- nursery hygiene
- crop rotation
- removal of infected plants
- use of disease-resistant varieties
- careful use of pesticides and fungicides

17. How different crops are affected

Maize

- pests damage leaves and cobs
- diseases reduce grain formation
- result: fewer and smaller cobs

Cotton

- pests damage bolls and leaves
- diseases reduce fibre quality
- result: low yield and poor quality

Tobacco

- pests and diseases damage leaves
- tobacco leaves are the main economic part
- result: reduced leaf quality and lower price

18. Importance of controlling pests and diseases

Controlling pests and diseases helps to:

- increase yields
- improve food security
- improve crop quality
- reduce losses
- increase farmer income
- protect the environment when safe methods are used

19. Practical/classroom discussion activities

Activity 1: Crop problem identification

Learners examine pictures of maize, cotton, or tobacco plants and identify:

- pest damage
- disease symptoms
- likely cause
- possible control method

Activity 2: Farm management plan

Learners design a control plan for one crop using:

- cultural methods
- chemical methods
- biological methods
- management methods

Activity 3: Symptom matching

Match symptoms such as:

- leaf holes
 - wilting
 - yellowing
 - rust spots
- with the likely pest or disease.

20. Summary

- Biomass is the organic content of a living organism.
- Productivity is the increase in biomass over time.
- Productivity is affected by light, mineral salts, temperature, water availability, pests and diseases.
- A pest is an organism that reduces productivity.
- Plant pests include tissue-eating and sap-sucking pests.
- Diseases such as bacterial wilt and fungal rust reduce yield.
- Pests and diseases can be controlled by management, cultural, chemical, and biological methods.
- Each method has advantages and disadvantages.
- Control methods should be chosen carefully for crops such as maize, cotton, and tobacco.

21. Key vocabulary

- biomass
- productivity
- pest
- disease
- yield
- infestation
- pathogen
- fungal rust
- bacterial wilt
- sap-sucking
- tissue-eating
- cultural control
- biological control
- chemical control
- management

22. Revision questions

Short-answer questions

1. Define biomass.
2. Define productivity.
3. State four factors affecting productivity.
4. What is a pest?
5. Name two types of plant pests.
6. Give two examples of plant diseases.
7. How do pests reduce productivity?
8. State two cultural methods of pest control.
9. Give one advantage and one disadvantage of chemical control.
10. Why is biological control useful?

Structured question

Describe how pests and diseases affect productivity in crops such as maize, cotton and tobacco.

Expected points

- damage leaves and reduce photosynthesis
- damage roots and reduce water absorption
- damage flowers/fruits/leaves
- lower yield and quality
- may kill plants early
- reduce farmer income

23. Board summary

- Biomass: organic content of living material.
- Productivity: increase in biomass over time.
- Factors affecting productivity: light, mineral salts, temperature, water, pests and diseases.
- Pests reduce productivity.
- Pests may be tissue-eating or sap-sucking.
- Diseases such as bacterial wilt and fungal rust reduce yields.
- Control methods: management, cultural, chemical and biological.
- Each method has strengths and weaknesses.
- Crops such as maize, cotton and tobacco need proper pest and disease management.

Transport in Plants

Learning objectives

By the end of the lesson, learners should be able to:

- describe the general structure of a plant
- distinguish between monocotyledonous and dicotyledonous plants
- identify parts of the internal structure of a young dicotyledonous stem and root
- compare the internal structure of dicotyledonous stem and root
- state the functions of xylem and phloem vessels
- define transpiration
- describe the functions of transpiration
- investigate factors affecting the rate of transpiration
- describe adaptations of leaves to minimize water loss
- describe how wilting occurs
- define translocation
- describe the effect of ring barking

1. Introduction

Plants need transport systems to move water, mineral salts, and manufactured food from one part of the plant to another. The main transport tissues are **xylem** and **phloem**. Leaves, stems, and roots are also adapted for transport and support.

2. General structure of a plant

A flowering plant has the following main parts:

- **Roots**
- **Stem**
- **Leaves**
- **Flowers**
- **Seeds**

Functions of the main parts

- **Roots:** anchor the plant, absorb water and mineral salts, and sometimes store food
- **Stem:** supports the plant and transports substances
- **Leaves:** make food by photosynthesis and carry out gaseous exchange and transpiration
- **Flowers:** reproductive organs
- **Seeds:** contain the embryo and food reserve for a new plant

3. Monocotyledonous and dicotyledonous plants

Monocotyledonous plants

These are plants with **one seed leaf** or **one cotyledon**.

Examples

- maize
- rice
- wheat
- grass

Main features

- one cotyledon
- leaves usually have **parallel venation**
- roots are usually **fibrous**
- vascular bundles are scattered in the stem
- flowers often have parts in multiples of three

Dicotyledonous plants

These are plants with **two seed leaves** or **two cotyledons**.

Examples

- bean
- sunflower
- hibiscus
- mango

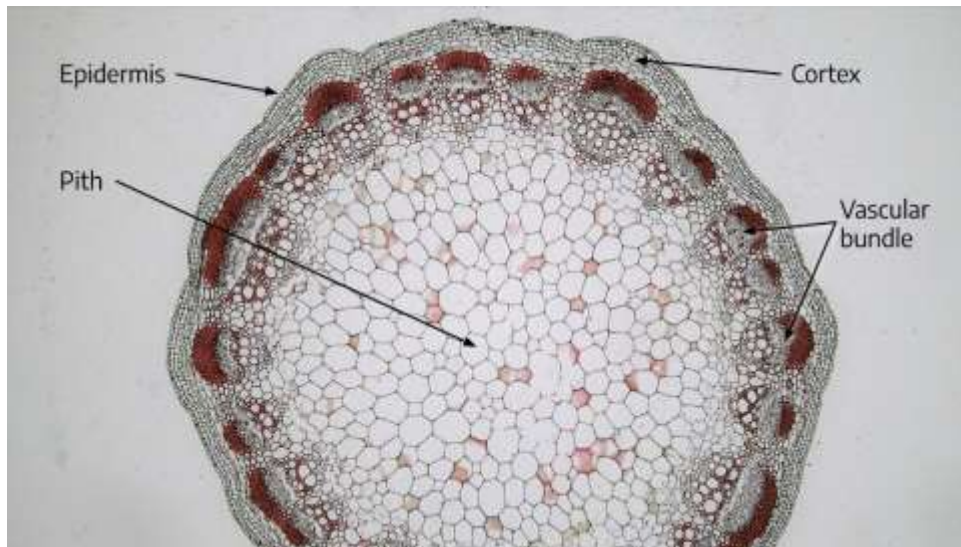
Main features

- two cotyledons
- leaves usually have **net-like venation**
- roots usually have a **tap root system**
- vascular bundles are arranged in a ring in the stem
- flowers often have parts in multiples of four or five

Differences between monocotyledonous and dicotyledonous plants

Feature	Monocotyledonous plants	Dicotyledonous plants
Cotyledons	One	Two
Leaf venation	Parallel	Net-like
Root system	Fibrous	Tap root
Stem vascular bundles	Scattered	In a ring
Flower parts	Multiples of 3	Multiples of 4 or 5

4. Internal structure of a young dicotyledonous stem



A young dicot stem has the following parts:

- **Epidermis**
- **Cortex**
- **Vascular tissue**
- **Pith**

a) Epidermis

The epidermis is the outermost layer.

Function

- protects the stem
- reduces water loss
- may have a waxy cuticle

b) Cortex

The cortex is found beneath the epidermis.

Function

- stores food
- provides support
- may contain chloroplasts in young stems

c) Vascular tissue

The vascular tissue is made of:

- **xylem**
- **phloem**

In a dicot stem, vascular bundles are arranged in a **ring**.

Function

- transport water, minerals, and food
- give support to the stem
-

d) Pith

The pith is found at the centre of the stem.

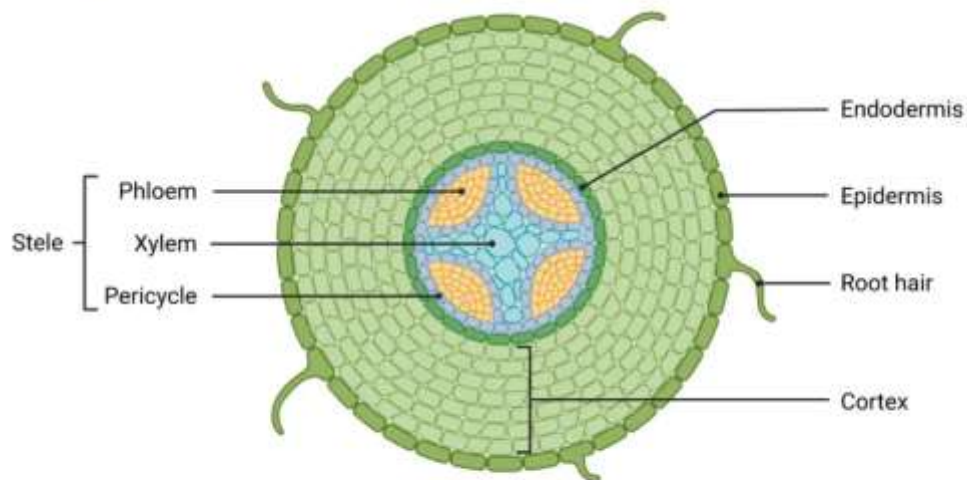
Function

- storage
- support in some plants

5. Internal structure of a young dicotyledonous root



Cross Section of a Dicot Root



A young dicot root has the following parts:

- **Epidermis**
- **Cortex**
- **Vascular tissue**
- **Root hairs**

a) Epidermis

The outer layer of the root.

Function

- protects the root
- allows root hair formation

b) Cortex

Located beneath the epidermis.

Function

- stores food
- allows passage of water to the centre of the root

c) Vascular tissue

Found at the centre of the root.

Function

- transports water and mineral salts through xylem
- transports food through phloem

d) Root hairs

Tiny outgrowths from the epidermis.

Function

- increase surface area for absorption of water and mineral salts

6. Comparison of dicot stem and dicot root

Feature	Dicot stem	Dicot root
Epidermis	Present	Present
Cortex	Present	Present
Vascular tissue	In a ring	At the centre
Pith	Usually present and large	Usually absent or very small
Root hairs	Absent	Present
Main role	Support and transport	Absorption and anchorage

7. Functions of xylem and phloem vessels

Xylem

Xylem vessels transport:

- water
- mineral salts

Other function

- provide support because their walls are thick and lignified

Direction of movement

- mainly upward from roots to leaves

Phloem

Phloem vessels transport:

- manufactured food, especially sugars such as sucrose

Direction of movement

- from leaves to other parts of the plant
- can move in both directions depending on where food is needed

8. Transpiration

Definition

Transpiration is the loss of water vapour from the aerial parts of a plant, especially through the stomata in the leaves.

Most water is lost through the **stomata**. Smaller amounts are lost through the cuticle and lenticels.

9. Functions of transpiration

Transpiration is useful to the plant in several ways.

a) Water movement in xylem

The loss of water from leaves creates a pulling force that helps water move up the xylem.

b) Movement of mineral ions

Transpiration helps mineral salts move from roots to leaves with the water stream.

c) Cooling effect

As water evaporates from the leaf surface, the plant cools down.

Importance

- prevents overheating
- helps maintain a suitable temperature for enzymes

10. Factors affecting the rate of transpiration

a) Surface area

A large leaf surface area increases water loss.

Effect

- larger surface area → faster transpiration
- smaller surface area → slower transpiration

b) Stomata distribution

Leaves with many stomata lose more water.

Effect

- more stomata → faster transpiration
- fewer stomata → slower transpiration

c) Temperature

High temperature increases evaporation.

Effect

- high temperature → faster transpiration
- low temperature → slower transpiration

d) Wind

Wind removes water vapour from around the leaf.

Effect

- strong wind → faster transpiration
- still air → slower transpiration

e) Humidity

Humidity is the amount of water vapour in the air.

Effect

- high humidity → slower transpiration
- low humidity → faster transpiration

f) Light intensity

Light causes stomata to open.

Effect

- high light intensity → stomata open wider → faster transpiration
- low light intensity → stomata close partly → slower transpiration

11. Investigating transpiration

Simple experiment using a potted plant

Materials

- potted plant
- transparent plastic bag
- Vaseline
- string or rubber band

Procedure

1. Cover the leaves of a potted plant with a transparent plastic bag.
2. Tie it around the stem to prevent outside air entering.
3. Leave the plant in a warm place.
4. Observe after some time.

Observation

Water droplets form inside the bag.

Explanation

The water droplets come from water vapour lost by transpiration.

Using Vaseline

Vaseline can be applied to leaf surfaces or stomata in experiments to reduce water loss and compare transpiration rates.

12. Adaptations of leaves to minimize water loss

Plants that live in dry environments have features that reduce transpiration.

a) Reduction of surface area

- leaves may be small, needle-like, or rolled
- reduces the area from which water is lost

b) Thick cuticle

- waxy layer reduces evaporation

c) Distribution of stomata

- stomata may be fewer
- stomata may be on the lower surface only
- helps reduce water loss

d) Presence of hairs

- hairs trap moist air near the leaf surface
- reduces evaporation

13. How wilting occurs

Definition

Wilting is the drooping of leaves and stems due to loss of water from the plant cells faster than water is absorbed.

How it happens

1. Water is lost by transpiration.
2. Roots cannot absorb water fast enough to replace the loss.
3. Cells lose turgor pressure.
4. Leaves and stems become limp and droop.

Significance of wilting

Wilting is a warning sign that the plant is losing too much water. It may:

- reduce photosynthesis
- reduce growth
- lead to death if water loss continues

14. Translocation

Definition

Translocation is the movement of manufactured food substances, especially sucrose, from leaves to other parts of the plant through the phloem.

What is transported?

- sucrose
- amino acids
- other organic nutrients

Where does it go?

- roots
- stems
- fruits
- seeds
- growing points

15. Importance of translocation

Translocation helps to:

- supply food to growing tissues
- store food in roots, tubers, and seeds
- support respiration and growth in non-photosynthetic parts of the plant
-

16. Effect of ring barking

Definition

Ring barking is the removal of a complete ring of bark from a stem.

The bark contains the **phloem**. When ring barking is done:

- phloem is removed
- translocation of food is stopped
- xylem may still carry water upward

Effects on the plant

- food cannot move from leaves to roots
- food accumulates above the ring
- roots may starve and die
- the plant may eventually die

Effect on the environment

- if many plants are ring barked, it can reduce plant survival and affect the ecosystem
- dead plants may reduce habitat and food for animals

17. Comparing xylem and phloem

Feature	Xylem	Phloem
Substance transported	Water and mineral salts	Manufactured food
Direction	Mainly upward	Both directions
Structure	Thick, lignified walls	Living cells with sieve tubes
Additional role	Support	Transport of food

18. Practical activities

Activity 1: Observing leaves

Learners observe leaves and identify:

- shape
- surface area
- venation
- stomata distribution
- presence of hairs

Activity 2: Transpiration experiment

Use a potted plant with a transparent bag to observe water droplets.

Activity 3: Ring barking demonstration

Use a stem diagram or model to show how removing bark affects food transport.

Activity 4: Internal structure observation

Use prepared diagrams or microscope slides of stems and roots to identify:

- epidermis
- cortex
- vascular tissue
- pith
- root hairs

19. Summary

- Plants have roots, stems, leaves, flowers and seeds.
- Monocots have one cotyledon; dicots have two.
- Dicot stems have vascular bundles in a ring; dicot roots have vascular tissue in the centre.
- Xylem transports water and minerals and gives support.
- Phloem transports manufactured food.
- Transpiration is the loss of water vapour from leaves.
- Transpiration helps in water movement, mineral movement and cooling.
- Factors affecting transpiration include surface area, stomata, temperature, wind, humidity and light.
- Leaves have adaptations that reduce water loss.
- Wilting happens when water loss exceeds water uptake.
- Translocation is the movement of food through phloem.
- Ring barking stops translocation and may kill the plant.

20. Key vocabulary

- xylem
- phloem
- transpiration
- translocation
- wilting
- monocotyledon
- dicotyledon
- root hair
- vascular tissue
- pith
- cortex
- epidermis
- stomata
- cuticle
- ring barking

21. Revision questions

Short-answer questions

1. State the main parts of a flowering plant.
2. Give two differences between monocotyledonous and dicotyledonous plants.
3. Name the tissues found in a young dicot stem.
4. Where is the vascular tissue found in a dicot root?
5. State the function of xylem.
6. State the function of phloem.
7. Define transpiration.
8. Give three factors that affect transpiration.
9. How do leaves reduce water loss?
10. Define translocation.
11. What happens when a plant is ring barked?

Structured question

Compare the internal structure of a young dicot stem and a young dicot root.

Expected answer points

- both have epidermis and cortex
- stem has vascular bundles in a ring and usually a pith
- root has vascular tissue at the centre and root hairs
- stem is mainly for support and transport, root for absorption and anchorage

22. Board summary

- Plant structure: roots, stem, leaves, flowers, seeds
- Monocots have one cotyledon; dicots have two
- Dicot stem: epidermis, cortex, vascular tissue, pith
- Dicot root: epidermis, cortex, vascular tissue, root hairs
- Xylem transports water and minerals; phloem transports food
- Transpiration is water loss from leaves
- Transpiration helps water movement and cooling
- Leaves have adaptations to reduce water loss
- Wilting occurs when cells lose turgor
- Translocation is food movement through phloem
- Ring barking stops translocation and may kill the plant

ANIMAL SCIENCE

Nutrition and Digestive System

Learning objectives

By the end of the lesson, learners should be able to:

- define a balanced diet
- list the components of a balanced diet
- list major sources of nutrients
- describe the major functions of nutrients
- test food for the presence of nutrients
- compare the energy content of carbohydrates and lipids
- relate food intake to age, sex, activity, and health requirements
- define malnutrition
- discuss causes and effects of malnutrition
- identify the main regions of the human alimentary canal and associated organs
- state functions of the parts of the alimentary canal
- define mechanical digestion
- describe peristalsis
- explain the importance of chemical digestion
- describe the functions of amylase, protease, and lipase
- explain the roles of hydrochloric acid, rennin, and bile
- describe structural adaptations of the small intestine to absorption
- describe absorption of end products of digestion
- state the function of the hepatic portal vein
- state the roles of the large intestine
- define assimilation
- describe types and functions of teeth
- describe the structure of a tooth
- identify causes of tooth decay
- describe the proper care of teeth
- outline the role of the liver in metabolism

1. Nutrition

1.1 Meaning of nutrition

Nutrition is the process by which living organisms take in food and use it for energy, growth, repair, and maintenance of the body.

Animals depend on other organisms for food, so they are called **heterotrophs**.

1.2 Balanced diet

Definition

A **balanced diet** is food that contains all the nutrients in the correct amounts and proportions needed by the body.

A balanced diet must supply:

- carbohydrates
- proteins
- fats and oils
- vitamins
- mineral salts
- water
- roughage

1.3 Components of a balanced diet and their functions

Nutrient	Major sources	Main function
Carbohydrates	Maize, rice, bread, potatoes, cassava, sugar	Main source of energy
Proteins	Meat, fish, eggs, beans, milk, peas, groundnuts	Growth and repair of tissues
Lipids	Cooking oil, butter, margarine, avocado, nuts	Energy storage, insulation, protection of organs
Vitamins	Fruits, vegetables, liver, milk, eggs	Protection against deficiency diseases and proper body functioning
Mineral salts	Milk, green vegetables, meat, salt, fish, iodised salt	Building bones, teeth, blood, and body functions
Water	Drinking water, fruits, vegetables, soups	Transport, digestion, excretion, and temperature control
Roughage	Vegetables, fruits, whole grains	Helps bowel movement and prevents constipation

1.4 Major functions of nutrients

Carbohydrates

- provide energy
- are stored as glycogen in animals

Proteins

- build body tissues
- repair worn-out tissues
- form enzymes, hormones, and antibodies

Lipids

- provide concentrated energy
- store energy
- reduce heat loss
- cushion organs

Vitamins

- help in normal body processes
- prevent deficiency diseases

Mineral salts

- calcium forms bones and teeth
- iron forms haemoglobin
- iodine is needed for the thyroid gland
- phosphorus helps in energy transfer and nucleic acids
- potassium helps ion and osmotic balance

Water

- medium for chemical reactions
- transport of substances
- cooling of the body
- excretion

Roughage

- adds bulk to food
- helps peristalsis
- prevents constipation

2. Food tests

2.1 Test for reducing sugars

Reagent

Benedict's solution

Procedure

1. Put food sample in a test tube.
2. Add Benedict's solution.
3. Heat in a hot water bath.

Positive result

- blue changes to green, yellow, orange, or brick-red precipitate

Example of reducing sugars

- glucose
- maltose

2.2 Test for non-reducing sugars

Reagents

- dilute hydrochloric acid
- sodium hydrogencarbonate or sodium hydroxide
- Benedict's solution

Procedure

1. Test the sample first with Benedict's solution.
2. If negative, boil with dilute acid.
3. Cool and neutralise.
4. Add Benedict's solution again and heat.

Positive result

- colour changes from blue to green/yellow/orange/brick-red

Example

- sucrose

2.3 Test for starch**Reagent**

Iodine solution

Positive result

- brown/yellow-brown to blue-black

2.4 Test for protein**Reagent**

Biuret reagent

Positive result

- blue to purple/violet/lilac

2.5 Test for lipids**Reagent**

Ethanol and water

Procedure

1. Shake food sample with ethanol.
2. Pour into water.

Positive result

- white/milky emulsion

2.6 Food test summary table

Nutrient	Test	Positive result
Reducing sugar	Benedict's + heat	Brick-red/orange precipitate
Non-reducing sugar	Hydrolysis + Benedict's	Brick-red/orange precipitate
Starch	Iodine	Blue-black
Protein	Biuret	Purple/violet
Lipid	Ethanol emulsion	Milky white emulsion

3. Energy content of carbohydrates and lipids

3.1 Comparison

- **Lipids** provide more energy per gram than carbohydrates.
- **Carbohydrates** provide less energy per gram but are easier and faster to use.

Key point

Lipids are more concentrated energy stores because they contain more energy in the same mass.

3.2 Importance

- carbohydrates are used for quick energy
- lipids are used for long-term energy storage

4. Nutrient requirements

Food intake should match the body's needs.

4.1 Age

- children need more nutrients for growth
- adults need nutrients for maintenance
- elderly people need less energy but still need balanced nutrition

4.2 Sex

- males often need more energy because of larger body size and higher muscle mass
- females need more iron, especially during menstruation and pregnancy

4.3 Activity

- active people need more energy
- athletes need more carbohydrates and water

4.4 Health condition

- sick people may need soft, nutritious food
- pregnant women need more proteins, iron, calcium, and vitamins
- lactating mothers need extra food for milk production

5. Malnutrition

5.1 Definition

Malnutrition is a condition caused by taking in too much or too little of one or more nutrients.

Malnutrition includes:

- **undernourishment**
- **overnourishment**

5.2 Undernourishment

This means not getting enough nutrients.

Effects of deficiency of important nutrients

Nutrient lacking	Deficiency signs
Vitamin A	Poor vision, night blindness
Vitamin C	Bleeding gums, scurvy
Vitamin D	Soft bones, rickets
Iron	Anaemia, tiredness
Calcium	Weak bones and teeth
Iodine	Swollen neck, goitre
Protein	Stunted growth, swelling, poor repair

5.3 Overnourishment

This means taking in too much food, especially energy-rich food such as fats and carbohydrates.

Effects

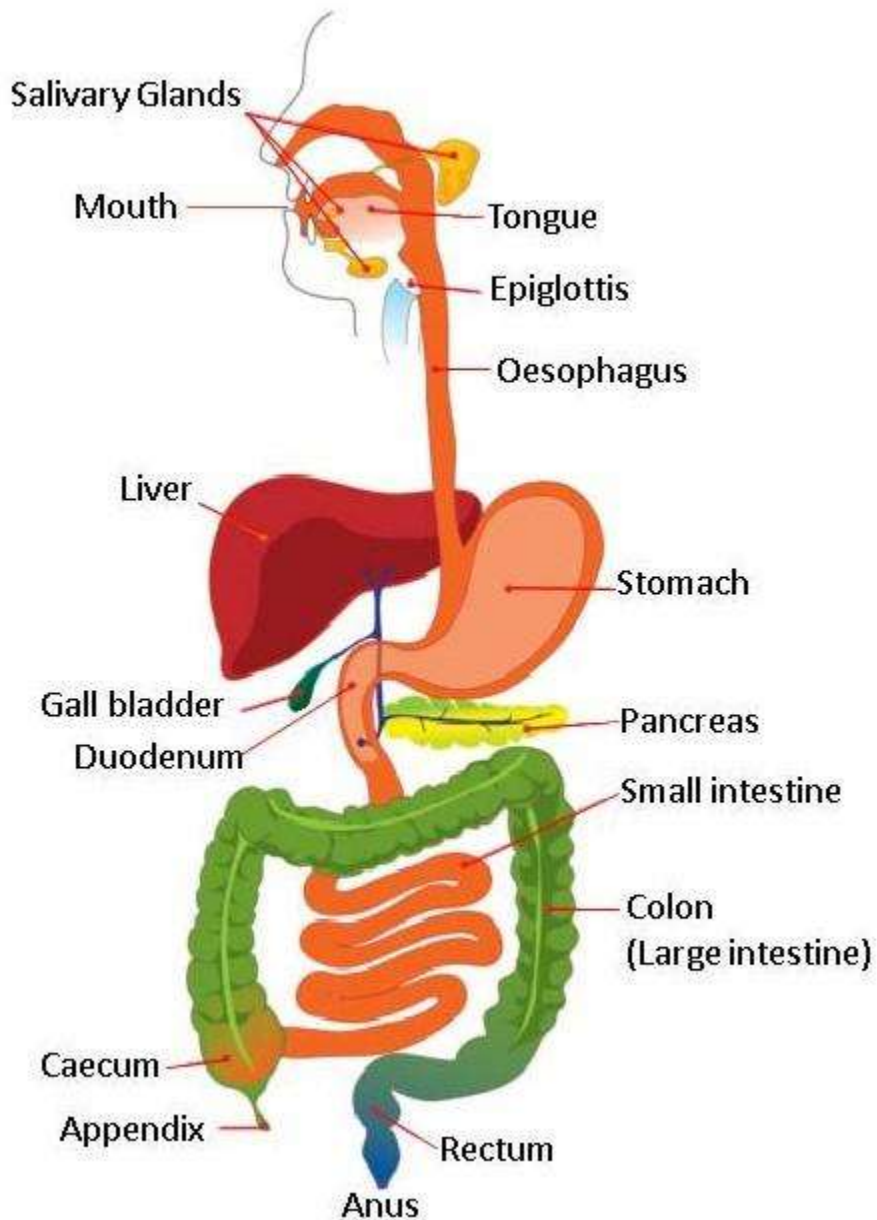
- obesity
- heart problems
- diabetes risk
- high blood pressure
- poor body movement and fatigue

5.4 Causes of malnutrition

- poor diet
- poverty
- ignorance about nutrition
- illness
- cultural food taboos
- overeating

- eating junk food instead of balanced meals

5. Human alimentary canal



6.1 Main regions

The human alimentary canal is a tube through which food passes.

Main parts

- mouth
- oesophagus
- stomach
- small intestine
- large intestine
- rectum
- anus

Associated organs include:

- salivary glands
- liver
- pancreas
- gall bladder

6.2 Functions of parts of the alimentary canal

Mouth

- ingestion of food
- chewing
- mixing with saliva
- starts starch digestion

Oesophagus

- moves food to the stomach by peristalsis

Stomach

- stores food temporarily
- churns food
- begins protein digestion
- kills bacteria with hydrochloric acid

Small intestine

- digestion continues
- absorption of digested food occurs here

Large intestine

- absorbs water and some vitamins
- forms faeces

Rectum

- stores faeces

Anus

- expels faeces

7. Digestive processes

The main processes of nutrition in animals are:

- ingestion
- digestion
- absorption
- assimilation
- egestion

7.1 Ingestion

Taking food into the body through the mouth.

7.2 Digestion

Breaking down large insoluble food molecules into small soluble molecules.

7.3 Absorption

Movement of digested food from the intestine into the blood or lymph.

7.4 Assimilation

Use of absorbed nutrients by body cells.

7.5 Egestion

Removal of undigested food as faeces through the anus.

8. Mechanical digestion and peristalsis

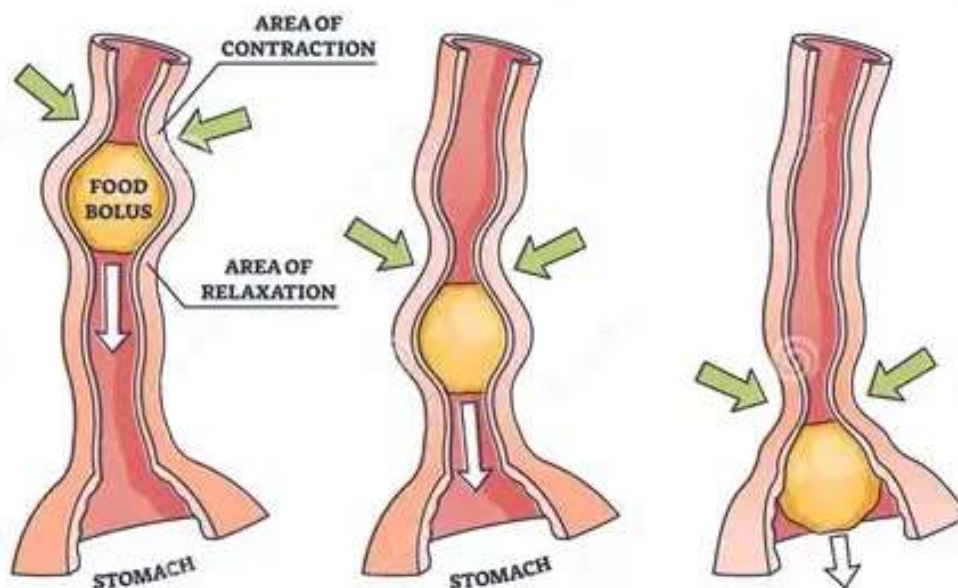
8.1 Mechanical digestion

Mechanical digestion is the physical breakdown of food into smaller pieces without chemical change.

Examples

- chewing by teeth
- churning in the stomach
- muscular movement of the gut

8.2 Peristalsis



Peristalsis is the wave-like contraction and relaxation of muscles in the wall of the alimentary canal that pushes food along.

Importance

- moves food from mouth to anus
- mixes food with digestive juices

9. Chemical digestion

Importance

Chemical digestion breaks large food molecules into small soluble molecules that can be absorbed.

Large molecules such as starch, proteins, and fats cannot pass through the intestinal wall in their original form.

9.1 Digestive enzymes

Enzyme	Acts on	Breaks down into	Site
Amylase	Starch	Maltose	Mouth, pancreas, small intestine
Protease	Proteins	Amino acids	Stomach, pancreas, small intestine
Lipase	Fats	Fatty acids and glycerol	Small intestine

9.2 Functions of specific digestive substances

Amylase

- converts starch to maltose

Protease

- converts proteins to amino acids

Lipase

- converts fats to fatty acids and glycerol

9.3 Role of hydrochloric acid

Hydrochloric acid is found in the stomach.

Functions

- provides acidic conditions for pepsin
- kills bacteria in food
- helps to break down food and activate enzymes

9.4 Role of rennin

Rennin is found in the stomach of young mammals.

Function

- clotting milk so that it stays longer in the stomach for digestion

9.5 Role of bile

Bile is made by the liver and stored in the gall bladder.

Functions

- emulsifies fats into small droplets
- neutralises acidic food from the stomach
- provides alkaline conditions for intestinal enzymes

10. Absorption in the small intestine

10.1 Structural adaptations of the small intestine

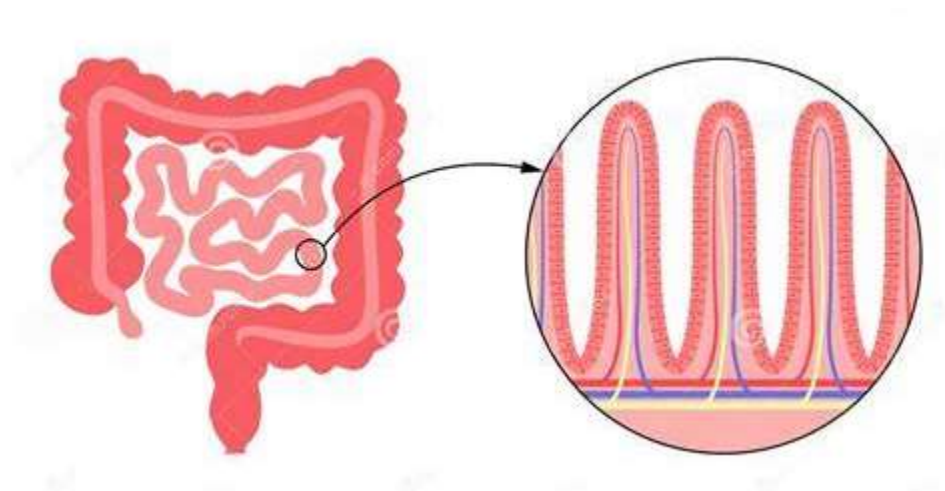
The small intestine is adapted for absorption because it has:

- a long length
- many folds
- villi
- thin lining
- rich blood supply

Function of each feature

- **long length:** more time and space for absorption
- **folds:** increase surface area
- **villi:** greatly increase surface area
- **thin lining:** short diffusion distance
- **rich blood supply:** carries away absorbed food quickly

10.2 Villi



Villi are finger-like projections in the small intestine.

Features of a villus

- thin epithelium
- many blood capillaries
- lacteal for absorption of fats
- large surface area

10.3 Absorption of end products of digestion

End products of digestion include:

- glucose
- amino acids
- fatty acids
- glycerol

How they are absorbed

- **glucose and amino acids** enter blood capillaries by diffusion and active transport
- **fatty acids and glycerol** enter lacteals

11. Hepatic portal vein

Function

The hepatic portal vein carries absorbed nutrients from the small intestine to the liver.

Importance

- food is processed before entering general circulation
- the liver can regulate blood sugar and amino acid levels

12. Large intestine

Roles of the large intestine

- absorption of water
- absorption of some vitamins, especially vitamin B formed by bacteria
- formation and storage of faeces
- expulsion of faeces through the anus

13. Assimilation

Definition

Assimilation is the use of absorbed food substances by body cells.

Examples

- glucose is used in respiration
- amino acids are used to make proteins
- fatty acids and glycerol are used to build fats
- excess glucose is stored as glycogen
- excess nutrients can be stored in the body

14. Role of the liver in metabolism

The liver is a very important organ in digestion and body metabolism.

Main roles

- regulates glucose concentration in blood
- converts excess glucose into glycogen
- converts glycogen back into glucose when needed
- deaminates excess amino acids
- breaks down alcohol and toxins
- stores fat, glycogen, vitamin A and vitamin D

Note

Detailed urea formation is not required here.

15. Teeth

15.1 Types of teeth and their functions

Type of tooth	Function
Incisors	Cutting food
Canines	Tearing food
Premolars	Crushing and grinding food
Molars	Grinding and chewing food

15.2 Structure of a tooth



External parts

- crown
- neck
- root

Internal parts

- enamel
- dentine
- pulp cavity
- blood vessels
- nerves
- cement
- root canal

15.3 Functions of tooth parts

Enamel

- hardest part
- protects the tooth

Dentine

- forms the main body of the tooth
- less hard than enamel

Pulp cavity

- contains blood vessels and nerves

Cement

- holds the tooth in the jaw

Root

- anchors tooth in jawbone

15.4 Comparison of external and internal structures

- the **crown** is visible above the gum
- the **root** is hidden in the jaw
- enamel covers the crown
- cement covers the root
- pulp contains living tissue and nerves

16. Tooth decay

16.1 Causes of tooth decay

Tooth decay is caused by bacteria acting on sugary food and producing acids.

How it happens

1. Food particles remain on teeth.
2. Bacteria feed on sugars.
3. Bacteria produce acids.
4. Acids dissolve enamel.
5. A cavity forms in the tooth.

16.2 Effects of tooth decay

- pain
- holes in teeth
- infection
- difficulty chewing
- loss of teeth if severe

17. Care of teeth

Proper care of teeth

- brush teeth at least twice a day
- use fluoride toothpaste
- remove plaque regularly
- rinse mouth after meals
- reduce sugary foods and drinks
- visit a dentist regularly

- use dental floss if necessary

Why care of teeth is important

- removes plaque
- neutralises acids
- destroys bacteria
- prevents decay and gum disease

18. Ruminants

Ruminants are herbivorous animals that digest plant material with the help of a specialised stomach.

Examples

- cow
- goat
- sheep
- deer

Main feature

They have a stomach with four compartments:

- rumen
- reticulum
- omasum
- abomasum

Importance

- helps digestion of cellulose-rich plant material
- contains microorganisms that break down fibre

19. Summary

Animals need a balanced diet for growth, energy, repair, and health. Nutrients include carbohydrates, proteins, lipids, vitamins, mineral salts, water, and roughage. Food tests help identify nutrients. The alimentary canal digests food mechanically and chemically, and the small intestine absorbs digested food. The liver controls metabolism and the large intestine absorbs water. Teeth help with mechanical digestion, and proper dental care prevents tooth decay. Ruminants have special digestive systems for plant foods.

20. Key vocabulary

- balanced diet
- malnutrition
- undernourishment
- overnourishment
- ingestion
- digestion
- absorption
- assimilation
- egestion
- peristalsis
- mechanical digestion
- chemical digestion
- enzyme
- villus
- hepatic portal vein
- metabolism
- tooth decay
- ruminant

21. Revision questions

Short-answer questions

1. Define a balanced diet.
2. State the main components of a balanced diet.
3. What is malnutrition?
4. Give two causes of malnutrition.
5. Name the parts of the human alimentary canal.
6. Define peristalsis.
7. What is mechanical digestion?
8. State the functions of amylase, protease, and lipase.
9. What is the function of bile?
10. Why are villi important?
11. Define assimilation.
12. State the function of the hepatic portal vein.
13. Give two causes of tooth decay.
14. How can tooth decay be prevented?

Structured question

Describe the digestion and absorption of food in humans.

Expected answer points

- food enters mouth by ingestion
- teeth and muscles carry out mechanical digestion
- salivary amylase starts starch digestion
- stomach churns food and uses acid and enzymes
- small intestine completes digestion
- bile emulsifies fats
- villi absorb end products
- glucose and amino acids enter blood
- fats enter lacteals
- liver receives nutrients through the hepatic portal vein

Board summary

- A balanced diet contains all nutrients in correct amounts.
- Nutrients: carbohydrates, proteins, lipids, vitamins, mineral salts, water, roughage.
- Malnutrition includes undernourishment and overnourishment.
- The alimentary canal includes mouth, oesophagus, stomach, small intestine, large intestine, rectum, anus.
- Digestion involves ingestion, digestion, absorption, assimilation, and egestion.
- Amylase digests starch, protease digests proteins, lipase digests fats.
- Bile emulsifies fats and neutralises acid.
- Villi increase absorption in the small intestine.
- The liver regulates nutrients and metabolism.
- Teeth are adapted for cutting, tearing, and grinding.
- Tooth decay is caused by bacteria on sugary food producing acids.
- Proper tooth care prevents decay.

Gaseous Exchange in Mammals

Learning objectives

By the end of the lesson, learners should be able to:

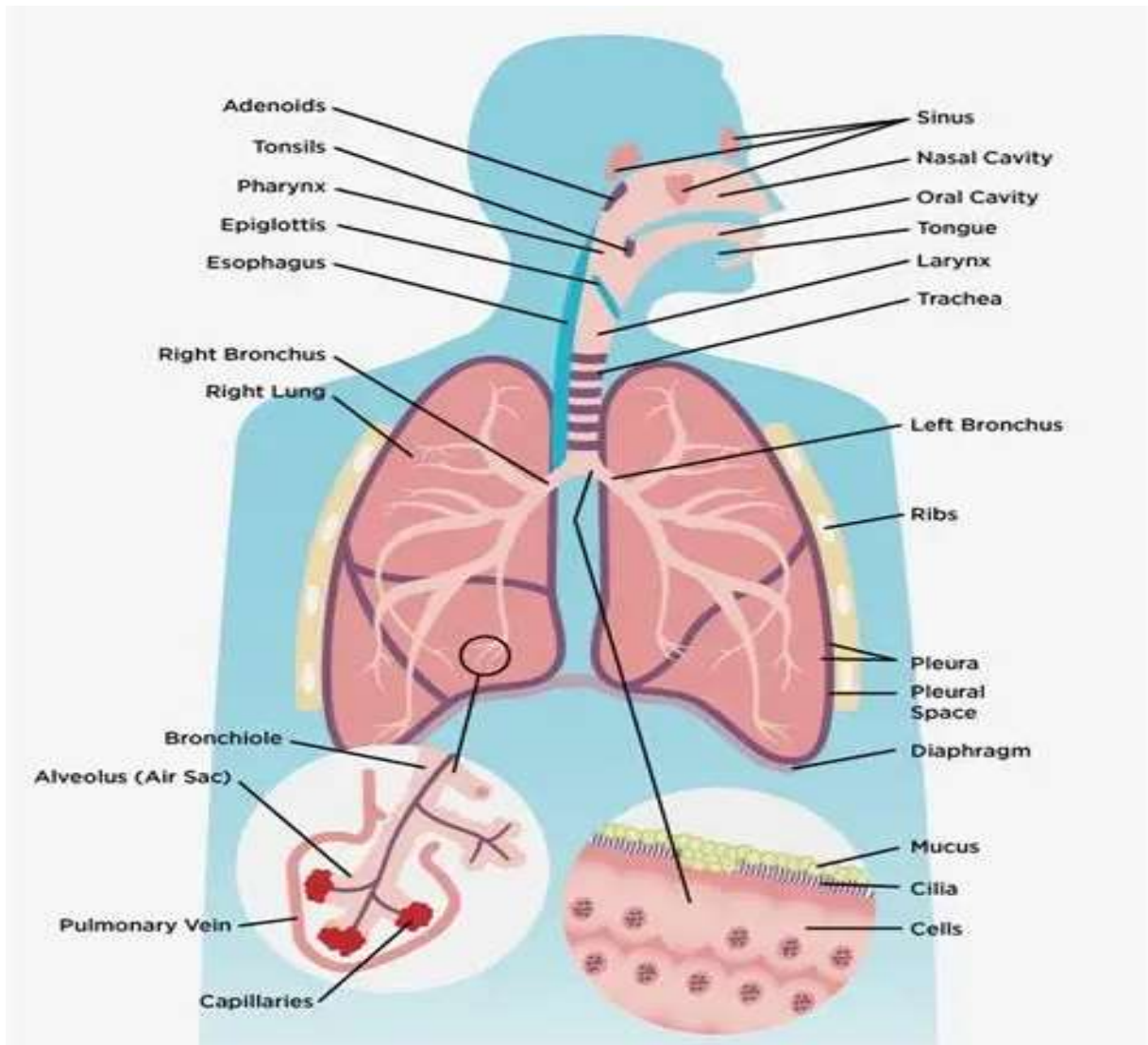
- identify parts of the respiratory system
- state the functions of parts of the respiratory system
- describe the mechanisms of breathing
- describe the structure and role of the alveoli
- describe factors which increase the efficiency of gaseous exchange
- state the difference between inhaled and exhaled air
- explain the effects of physical activity on breathing and blood circulation
- describe some effects of smoking on the respiratory system

1. Introduction

Mammals need oxygen for respiration and must remove carbon dioxide produced during respiration. Gaseous exchange is the movement of oxygen into the body and carbon dioxide out of the body.

In mammals, gaseous exchange takes place mainly in the **alveoli** of the lungs.

3. Parts of the respiratory system



The main parts of the respiratory system are:

- nasal passages
- larynx
- trachea
- bronchi
- bronchioles
- alveoli
- capillaries
- lungs
- diaphragm
- intercostal muscles

3. Functions of the parts of the respiratory system

a) Nasal passages

- air enters the body through the nose
- they warm the air
- they moisten the air
- they filter dust and germs using hairs and mucus

b) Larynx

- also called the voice box
- contains the vocal cords
- allows sound production
- directs air into the trachea

c) Trachea

- carries air from the larynx to the bronchi
- supported by rings of cartilage to keep it open
- lined with mucus and cilia

d) Bronchi

- two branches of the trachea
- each bronchus carries air to one lung

e) Bronchioles

- smaller branches of the bronchi
- carry air to the alveoli

f) Alveoli

- tiny air sacs where gaseous exchange takes place
- oxygen diffuses into the blood
- carbon dioxide diffuses out of the blood

g) Capillaries

- tiny blood vessels surrounding the alveoli
- provide a good blood supply for gas exchange

h) Lungs

- contain the respiratory surfaces
- protected by the rib cage
- expand and contract during breathing

i) Diaphragm

- dome-shaped muscle beneath the lungs
- helps bring about breathing movements

j) Intercostal muscles

- muscles between the ribs
- move the rib cage during breathing

4. Mechanism of breathing

Breathing involves two main processes:

- inhalation
- exhalation

4.1 Inhalation

Inhalation is the intake of air into the lungs.

What happens during inhalation

- the diaphragm contracts and becomes flat
- the external intercostal muscles contract
- the ribs move upward and outward
- the chest cavity becomes larger
- pressure inside the chest cavity decreases
- air enters the lungs

Result

The lungs expand and air is drawn in.

4.2 Exhalation

Exhalation is the removal of air from the lungs.

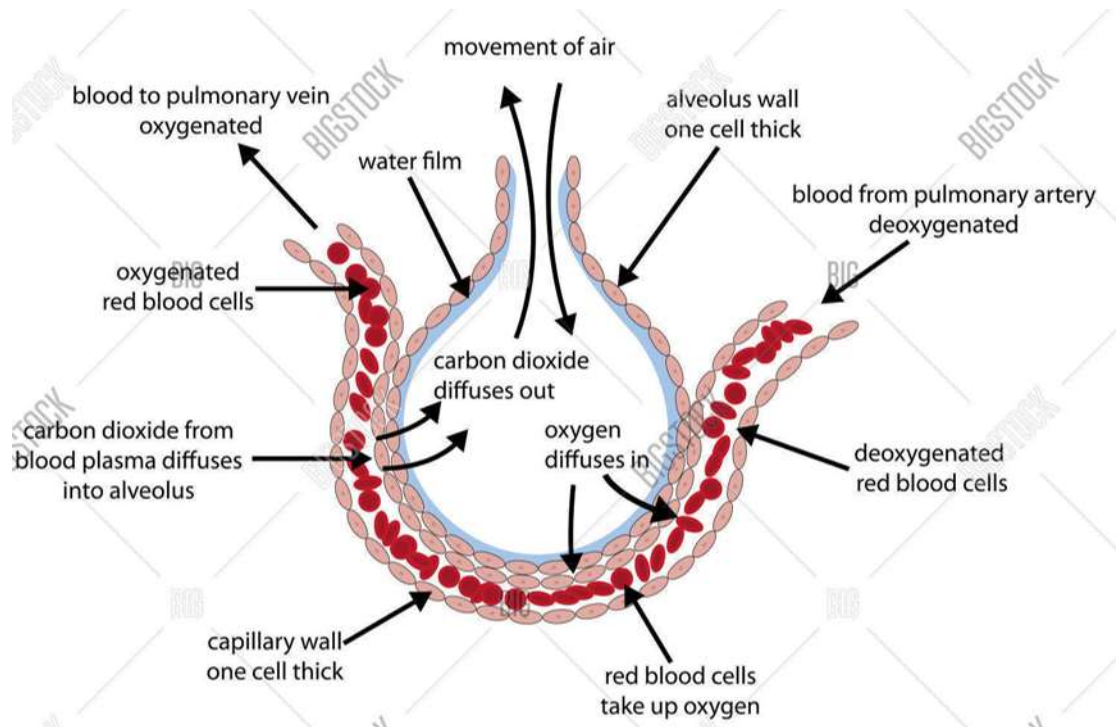
What happens during exhalation

- the diaphragm relaxes and becomes dome-shaped
- the intercostal muscles relax
- the ribs move downward and inward
- the chest cavity becomes smaller
- pressure in the chest cavity increases
- air is forced out of the lungs

Result

The lungs shrink and air leaves the body.

5. Structure and role of the alveoli



5.1 Structure of alveoli

Alveoli are tiny air sacs with the following features:

- very large surface area
- thin walls, only one cell thick
- moist lining
- surrounded by many capillaries
- numerous in number

5.2 Role of alveoli

Alveoli are the site of gaseous exchange.

How gaseous exchange occurs

- oxygen diffuses from the alveoli into the blood
- carbon dioxide diffuses from the blood into the alveoli

This happens because of differences in concentration.

6. Factors that increase the efficiency of gaseous exchange

Gaseous exchange is efficient because the alveoli have adaptations that make diffusion fast.

a) Large surface area

- many alveoli provide a very large surface area
- more oxygen and carbon dioxide can diffuse at the same time

b) Thin walls

- alveolar walls are one cell thick
- this shortens the diffusion distance

c) Moist surface

- gases dissolve in moisture before diffusing
- this makes diffusion easier

d) Rich blood supply

- many capillaries surround each alveolus
- blood carries away oxygen and brings carbon dioxide
- maintains a steep diffusion gradient

e) Steep diffusion gradient

- oxygen concentration is high in the alveoli and low in blood
- carbon dioxide concentration is high in blood and low in alveoli
- this promotes rapid diffusion

7. Differences between inhaled and exhaled air

Component	Inhaled air	Exhaled air
Oxygen	More	Less
Carbon dioxide	Less	More
Water vapour	Less	More
Temperature	Lower	Higher
Nitrogen	About the same	About the same

Explanation

- oxygen is used in respiration
- carbon dioxide is produced during respiration
- air is warmed and moistened in the body before being breathed out

8. Effects of physical activity on breathing and blood circulation

During exercise, muscles need more oxygen and produce more carbon dioxide.

Effects on breathing

- breathing rate increases
- breathing becomes deeper
- more oxygen enters the body
- more carbon dioxide is removed

Effects on blood circulation

- pulse rate increases
- heart beats faster
- blood is pumped more quickly to muscles
- more oxygen and nutrients are delivered to active tissues

Why this happens

The body needs extra energy during physical activity, so respiration increases.

9. Smoking and the respiratory system

Smoking damages the respiratory system and reduces the efficiency of gaseous exchange.

9.1 Short-term effects of smoking

a) Destruction of cilia

- cilia line the air passages and help remove dust and mucus
- smoking destroys cilia
- dust and germs remain in the lungs

b) Increased mucus production

- smoke irritates the airways
- more mucus is produced
- mucus may block air passages

c) Constriction of bronchioles

- smoke causes narrowing of air passages
- breathing becomes difficult
- less air reaches the alveoli

9.2 Long-term effects of smoking

a) Bronchitis

- inflammation of the bronchi
- causes coughing, mucus build-up, and difficulty breathing

b) Lung cancer

- caused by harmful chemicals in tobacco smoke
- cancerous growths damage lung tissue
- may lead to death if untreated

Other possible long-term effects

- reduced lung capacity
- chronic coughing
- increased risk of pneumonia
- poor oxygen supply to the body

10. Why smoking is harmful

Smoking:

- damages the lungs
- reduces gaseous exchange
- increases risk of disease
- reduces physical fitness
- may shorten life

11. Demonstrations and practical activities

Activity 1: Breathing model

Use a bottle, balloons, and a rubber sheet to model the lungs and diaphragm.

What it shows

- inhalation
- exhalation
- movement of the diaphragm and chest cavity

Activity 2: Observation of breathing rate

Measure breathing rate before and after exercise.

What to observe

- breathing becomes faster after activity
- pulse rate also increases

Activity 3: Smoking and cilia discussion

Use diagrams or pictures to show how smoke damages the cilia and lungs.

12. Summary

- The respiratory system in mammals includes the nasal passages, larynx, trachea, bronchi, bronchioles, alveoli, capillaries, lungs, diaphragm, and intercostal muscles.
- Inhalation increases chest volume and draws air into the lungs.
- Exhalation decreases chest volume and forces air out.
- Alveoli are adapted for efficient gaseous exchange because they are numerous, thin, moist, and richly supplied with blood.
- Inhaled air has more oxygen and less carbon dioxide than exhaled air.
- Physical activity increases breathing rate, depth of breathing, and pulse rate.
- Smoking damages the respiratory system and may cause bronchitis and lung cancer.

13. Key vocabulary

- gaseous exchange
- respiratory system
- inhalation
- exhalation
- alveoli
- bronchioles
- diaphragm
- cilia
- mucus
- diffusion gradient
- pulse rate
- bronchitis
- lung cancer

14. Revision questions

Short-answer questions

1. Name the parts of the respiratory system.
2. State the function of the trachea.
3. What happens to the diaphragm during inhalation?
4. Why are alveoli suitable for gaseous exchange?
5. Give two differences between inhaled and exhaled air.
6. State two effects of exercise on breathing.
7. List three short-term effects of smoking.
8. List two long-term effects of smoking.

Structured question

Describe the mechanism of breathing in mammals.

Expected answer points

- inhalation: diaphragm contracts, ribs move up and out, chest cavity volume increases, pressure decreases, air enters
- exhalation: diaphragm relaxes, ribs move down and in, chest cavity volume decreases, pressure increases, air leaves

15. Board summary

- Respiratory system parts: nasal passages, larynx, trachea, bronchi, bronchioles, alveoli, capillaries, lungs, diaphragm, intercostal muscles.
- Inhalation increases chest volume; exhalation decreases chest volume.
- Alveoli have a large surface area, thin walls, moist lining, and rich blood supply.
- Inhaled air has more oxygen, less carbon dioxide, and less water vapour than exhaled air.
- Exercise increases breathing rate and pulse rate.
- Smoking damages cilia, increases mucus, constricts bronchioles, and can cause bronchitis and lung cancer.

Topic: Respiration

By the end of the lesson, learners should be able to:

- define aerobic respiration
- state the word equation for aerobic respiration
- list uses of energy in the body
- define anaerobic respiration
- state the word equations for anaerobic respiration in plants and animals
- describe the production and effects of lactic acid on muscles during exercise

1. Introduction

Living organisms need energy for all life processes. Energy is released from food during respiration. Respiration is not the same as breathing. Breathing is the movement of air in and out of the lungs, while respiration is a chemical process inside cells that releases energy.

2. Aerobic respiration

Definition

Aerobic respiration is the breakdown of food, especially glucose, in the presence of oxygen to release energy.

It takes place in living cells, mainly in the mitochondria.

3. Word equation for aerobic respiration

glucose + oxygen → carbon dioxide + water + energy

4. Importance of aerobic respiration

Aerobic respiration releases a large amount of energy, which is used by the body for many activities.

5. Uses of energy in the body

Energy from respiration is used for:

- muscle contraction and movement
- active transport
- growth and repair of tissues
- cell division
- synthesis of proteins and other substances
- transmission of nerve impulses
- maintaining body temperature in mammals
- pumping blood by the heart
- movement of cilia and flagella in some organisms

Key point

Without energy, cells cannot function properly and life processes would stop.

6. Anaerobic respiration

Definition

Anaerobic respiration is the breakdown of food, especially glucose, without oxygen to release a small amount of energy.

It happens when there is not enough oxygen available.

7. Word equations for anaerobic respiration

a) In plants and yeast

glucose → ethanol + carbon dioxide + energy

b) In animals

glucose → lactic acid + energy

8. Comparison of aerobic and anaerobic respiration

Feature	Aerobic respiration	Anaerobic respiration
Oxygen needed	Yes	No
Amount of energy released	Large	Small
End products	Carbon dioxide and water	Ethanol and carbon dioxide in plants; lactic acid in animals
Site in cell	Mainly mitochondria	In the cytoplasm
Efficiency	High	Low

9. Anaerobic respiration in plants

In plants and yeast, when oxygen is not available, glucose is broken down into ethanol and carbon dioxide.

Word equation

glucose → ethanol + carbon dioxide + energy

Importance

This process is useful in:

- bread making
- brewing
- production of alcoholic drinks and some industrial processes

Note

The process in yeast is called **fermentation**.

10. Anaerobic respiration in animals

In animals, especially in muscle cells during vigorous exercise, glucose is broken down into lactic acid.

Word equation

glucose → lactic acid + energy

Why it happens

During heavy exercise, muscles may need more oxygen than the body can supply. The muscles then respire anaerobically.

11. Lactic acid production during exercise

How lactic acid is formed

When muscles work hard and oxygen supply becomes insufficient:

- glucose is broken down without oxygen
- lactic acid is produced
- only a small amount of energy is released

Effects of lactic acid on muscles

- muscle fatigue
- muscle cramps
- pain in muscles
- reduced ability to continue exercising

Fatigue

Fatigue is the feeling of weakness and tiredness in muscles after prolonged activity.

12. Oxygen debt

Definition

Oxygen debt is the extra oxygen needed after vigorous exercise to break down the lactic acid produced during anaerobic respiration.

Why oxygen debt occurs

During exercise:

- some respiration in muscles becomes anaerobic
- lactic acid accumulates
- after exercise, extra oxygen is required to oxidise the lactic acid

What happens after exercise

The body:

- breathes faster and deeper
- pulse rate remains high for a while
- oxygen is taken to the muscles
- lactic acid is broken down

Result

The body repays the oxygen debt and the muscles recover.

13. Effects of strenuous exercise on respiration

During hard exercise:

- breathing rate increases
- breathing becomes deeper
- pulse rate increases
- more oxygen reaches the muscles
- more carbon dioxide is removed

If oxygen supply is still not enough:

- anaerobic respiration occurs
- lactic acid builds up
- fatigue and cramps may develop

14. Why aerobic respiration is important

Aerobic respiration is important because:

- it releases a lot of energy
- it provides energy for all body activities
- it supports efficient long-term activity
- it produces harmless waste products, carbon dioxide and water

15. Why anaerobic respiration is less efficient

Anaerobic respiration releases less energy because glucose is only partly broken down.

This means:

- less energy is available
- lactic acid may accumulate in animals
- the process is not suitable for long-term energy supply

16. Practical/classroom demonstration ideas

Activity 1: Breathing rate after exercise

Learners measure:

- pulse rate before exercise
- pulse rate after exercise
- breathing rate before and after exercise

Observation

Both pulse rate and breathing rate increase after exercise.

Explanation

The body needs more oxygen and energy during activity.

Activity 2: Discussion on lactic acid

Discuss how footballers, runners, or cyclists may experience tired muscles after intense exercise.

Activity 3: Yeast fermentation demonstration

Use yeast, sugar, and warm water to show gas production.

Observation

Bubbles of carbon dioxide are produced.

Explanation

Yeast is carrying out anaerobic respiration.

17. Common misconceptions

- Respiration is not the same as breathing.
- Anaerobic respiration does not mean “no energy”; it means less energy.
- Lactic acid is formed in animal muscles during oxygen shortage, not in plants.
- Carbon dioxide is produced in anaerobic respiration in plants and yeast, but not in animals.

18. Summary

Respiration is the process of releasing energy from food in living cells. Aerobic respiration uses oxygen and releases a large amount of energy, while anaerobic respiration occurs without oxygen and releases less energy. In plants and yeast, anaerobic respiration produces ethanol and carbon dioxide. In animals, it produces lactic acid. Lactic acid causes fatigue and oxygen debt during vigorous exercise.

19. Key vocabulary

- respiration
- aerobic respiration
- anaerobic respiration
- glucose
- oxygen
- energy
- lactic acid
- ethanol
- oxygen debt
- fatigue
- mitochondria
- fermentation

20. Revision questions

Short-answer questions

1. Define aerobic respiration.
2. Write the word equation for aerobic respiration.
3. State four uses of energy in the body.
4. Define anaerobic respiration.
5. Write the word equation for anaerobic respiration in plants.
6. Write the word equation for anaerobic respiration in animals.
7. What is lactic acid?
8. What is oxygen debt?
9. Explain why muscles feel tired after vigorous exercise.

Structured question

Compare aerobic and anaerobic respiration.

Expected answer points

- aerobic uses oxygen, anaerobic does not
- aerobic releases more energy, anaerobic less
- aerobic produces carbon dioxide and water, anaerobic produces ethanol and carbon dioxide in plants or lactic acid in animals
- aerobic is efficient, anaerobic is less efficient

21. Board summary

- Aerobic respiration: $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$
- Anaerobic respiration in plants: $\text{glucose} \rightarrow \text{ethanol} + \text{carbon dioxide} + \text{energy}$
- Anaerobic respiration in animals: $\text{glucose} \rightarrow \text{lactic acid} + \text{energy}$
- Energy is used for movement, growth, active transport, body temperature regulation, and other life processes
- Lactic acid causes fatigue
- Oxygen debt is the extra oxygen needed to break down lactic acid after exercise

Topic: Transport in Mammals

Learning objectives

By the end of the lesson, learners should be able to:

- describe the human circulatory system
- state the functions of the circulatory system
- identify parts of the heart
- describe the cardiac cycle
- explain the importance of heart valves
- relate pressure differences in the dual circulatory system to the functions of the two circuits
- relate the structure of blood vessels to their function
- identify possible causes of high blood pressure
- identify possible causes of coronary heart disease
- state the components of mammalian blood and their functions
- describe the movement of materials between capillaries and tissue fluid

1. Introduction

The transport system in mammals is responsible for moving materials around the body. It carries oxygen, nutrients, hormones, and waste products. It also helps distribute heat and plays a role in immunity.

The human circulatory system is made up of:

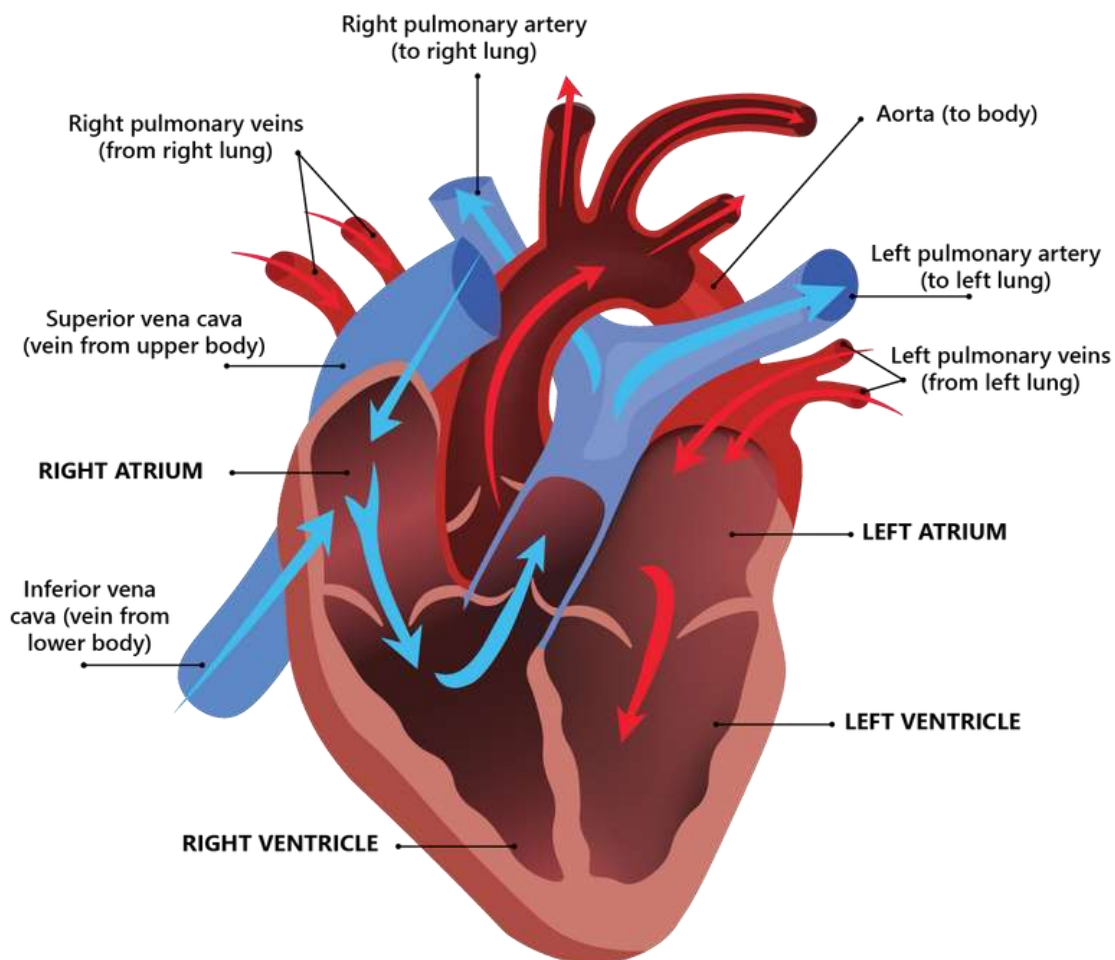
- the heart
- blood vessels
- blood
- valves

2. Functions of the circulatory system

The circulatory system performs the following functions:

- transports oxygen from the lungs to the body cells
- transports carbon dioxide from body cells to the lungs
- transports digested food from the small intestine to body cells
- transports hormones from glands to target organs
- carries waste products to excretory organs
- distributes heat around the body
- helps protect the body against disease through white blood cells and antibodies

3. The human heart



3.1 Parts of the heart

The heart is a muscular pump divided into four chambers:

- right atrium
- left atrium
- right ventricle
- left ventricle

Other important parts

- blood vessels entering and leaving the heart
- valves

3.2 Functions of the main parts of the heart

Atria/auricles

- receive blood returning to the heart

Ventricles

- pump blood out of the heart
- the left ventricle pumps blood to the whole body
- the right ventricle pumps blood to the lungs

Valves

- prevent blood from flowing backward
- ensure one-way movement of blood

Major blood vessels connected to the heart

- vena cava
- pulmonary artery
- pulmonary vein
- aorta

4. The cardiac cycle

The cardiac cycle is the sequence of events in one complete heartbeat.

Main events

1. The atria contract and push blood into the ventricles.
2. The ventricles contract and push blood into the arteries.
3. The heart muscles relax and the chambers fill with blood again.

Importance

This cycle ensures continuous movement of blood through the body and lungs.

5. Importance of heart valves

Heart valves are important because they:

- prevent backflow of blood
- maintain pressure needed for efficient circulation
- ensure blood moves in one direction only

Types of valves

- atrioventricular valves: between atria and ventricles
- semilunar valves: between ventricles and arteries

6. Double circulation

Definition

Double circulation means blood passes through the heart twice in one complete circulation of the body.

It has two circuits:

- pulmonary circulation
- systemic circulation

6.1 Pulmonary circulation

This is the movement of blood from the heart to the lungs and back to the heart.

Pathway

- right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium

Function

- carries deoxygenated blood to the lungs for oxygenation
- returns oxygenated blood to the heart

6.2 Systemic circulation

This is the movement of blood from the heart to the rest of the body and back to the heart.

Pathway

- left ventricle → aorta → body tissues → vena cava → right atrium

Function

- supplies oxygen and nutrients to body tissues
- removes waste products from tissues

6.3 Pressure differences in double circulation

The two circuits have different pressure needs:

Pulmonary circulation

- works at lower pressure
- because the blood only goes to the lungs, which are close to the heart
- low pressure protects the delicate lung capillaries

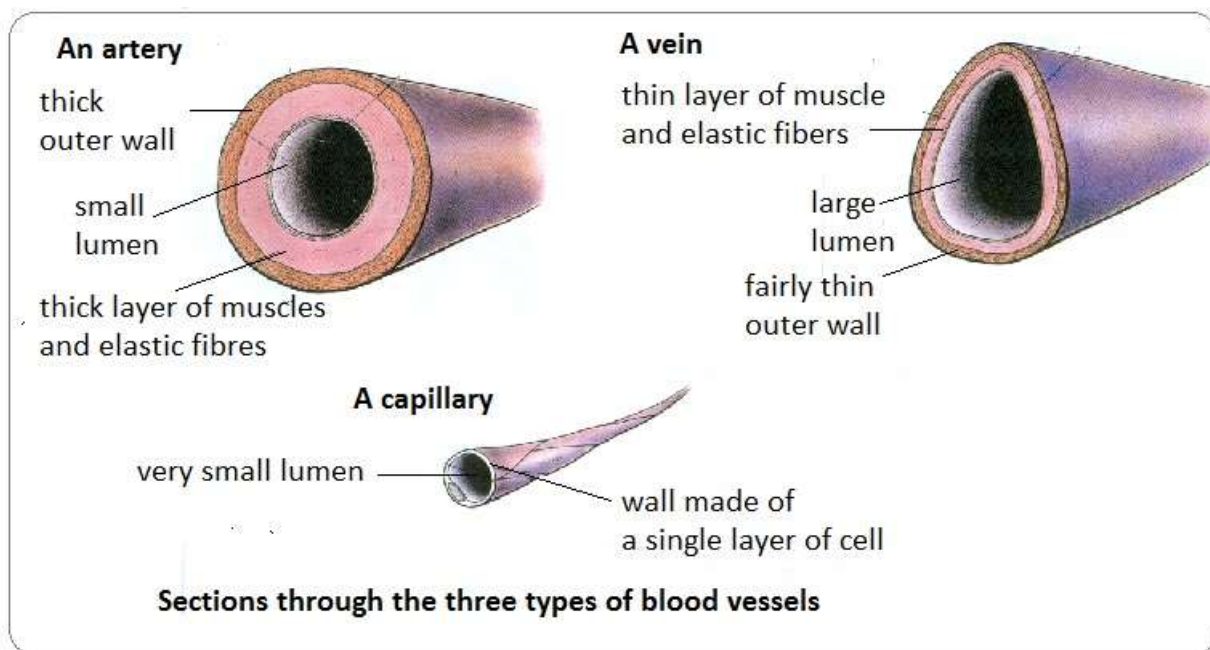
Systemic circulation

- works at higher pressure
- because blood must travel to the whole body
- high pressure ensures fast delivery of oxygen and nutrients

Importance

These pressure differences make blood flow efficiently in both circuits.

7. Blood vessels



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